

Formal Certification Methods for Automated Vehicle Safety Assessment

Tong Zhao^{ID}, *Member, IEEE*, Ekim Yurtsever^{ID}, *Member, IEEE*, Joel A. Paulson^{ID},
and Giorgio Rizzoni^{ID}, *Fellow, IEEE*

Abstract—Challenges related to automated driving are no longer focused on just the construction of such automated vehicles (AVs) but also on assuring the safety of operation. Recent advances in Level 3 and Level 4 autonomous driving have motivated more extensive study in safety guarantees of complicated AV maneuvers, which aligns with the goal of ISO 21448 (Safety of the Intended Functions or SOTIF), i.e., minimizing unsafe scenarios both known and unknown, as well as Vision Zero – eliminating highway fatalities by 2050. A majority of approaches used in providing safety guarantees for AV motion control originate from formal methods, especially reachability analysis (RA), which relies on mathematical models for the dynamic evolution of the system to provide guarantees. However, to the best of the authors' knowledge, there have been no review papers dedicated to describing and interpreting state-of-the-art of formal methods in the context of AVs. In this work, we provide both an overview of the safety verification, validation, and certification process, as well as a review of formal safety techniques that are best suited to AV applications. We also propose a unified scenario coverage framework that can provide either a formal or sample-based estimate of safety verification for full AVs. Finally, challenges and future opportunities beyond the scope of current published research for assured AV safety are presented.

Index Terms—Automated vehicle, verification and validation, reachability analysis, safety guarantees, motion planning, safety of the intended function, scenario coverage.

I. INTRODUCTION

AUTOMATED vehicle (AV) technology is expected to be massively deployed in the next decade or two, with multiple SAE J3016 Level 2 automated driving systems developed and deployed over the last two decades [1], [2], and some level 3 systems regionally approved and announced [3], [4]. As a daily means of transportation for humans and goods, more rigorous safety guarantees have to be investigated and eventually implemented in the automated fleet that may dominate the road

traffic around 2050 [5]. A recent poll suggests that safety is the primary concern for AV acceptance, compared to economic impact or privacy concerns [6]. Although some believe that the mass adoption of automated vehicles will significantly improve road safety [7], the gap between this promising autonomous future and the current state-of-the-art safety measures is still vast, requiring the continued effort of scientists, engineers, and legislators.

Previous surveys on AV technologies have been focusing on the various aspects of the technical challenges and implementation, such as ways to perform perception, decision making, vehicle control, human-machine interaction [8], with only a few reviews covering the issue of AV safety verification and validation [9]–[16].

With the rise of real-world deployment of ADS systems, a significant amount of verification and validation (V&V) is required against the scale of real-world scenarios that ADS will encounter. A common strategy of V&V to maximize scenario coverage is to verify ADS in virtual simulations and simulate a large amount of generated scenario samples (see [14], [17] for detailed reference).

The challenge in scenario sampling-based V&V is quantifying the amount of sample required to guarantee enough coverage, so that the risk created by improper ADS actuation can be curbed. A reasonable scenario coverage guarantee is also the prerequisite for type approval in recent legislations for highly automated vehicle (HAV) deployment [18], [19]. However, an alternative set of methods based on formal verification addresses the scenario coverage from a different angle through specification satisfaction [15]. Because of the potential of formal methods in scenario coverage, emerging research has started to combine formal properties with control synthesis [20], [21] and safety verification [22].

In light of the need for ADS quantifiable verification, we propose a unified scenario coverage framework to address the unresolved pain point in AV scenario coverage. A candidate AV control policy can go through either formal or sample-based safety verification to be tested for its safe scenario coverage in a specified ODD. We compare the pros and cons of sample-based methods and formal methods based on how they reach this safety coverage and the associated costs and issues with each method.

The contributions of this paper are three-fold: (1) we provide a holistic overview of different aspects of ADS safety guarantee, verification, validation, and certification; (2) we provide a unified safe scenario coverage framework for both sample-based

Manuscript received 10 March 2022; revised 6 April 2022; accepted 18 April 2022. Date of publication 4 May 2022; date of current version 23 January 2023. (Corresponding author: Tong Zhao.)

Tong Zhao and Giorgio Rizzoni are with the Department of Mechanical and Aerospace Engineering, The Ohio State University, Columbus, OH 43212 USA (e-mail: zhao.1991@osu.edu; rizzoni.1@osu.edu).

Ekim Yurtsever is with the Department of Electrical and Computer Engineering, The Ohio State University, Columbus, OH 43212 USA (e-mail: yurtsever.2@osu.edu).

Joel A. Paulson is with the Department of Chemical and Biomolecular Engineering, The Ohio State University, Columbus, OH 43212 USA (e-mail: paulson.82@osu.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIV.2022.3170517>.

Digital Object Identifier 10.1109/TIV.2022.3170517

TABLE I
LEGISLATION STATUS OF AUTOMATED VEHICLE OPERATION

Region	Testing	Deployment Legislation		Robotaxi Safety Testing	Driver Removal Deployment	Approved Deployment Case(s)
		Level 3 (L3)	Level 4 (L4)			
US	✓*	State-specific** (J3016 classification[23] in 2014)	Steering wheel & pedal delete, Mar 2022[24]	Conditional***	Pending****	Nuro (CA, Dec 2020)[25] Waymo, Cruise (CA, Sep 2021)[26]
EU	✓	UNECE No.79, Apr 2021[27] Germany, Jun 2017[18]	Germany, Jul 2021[28]	Germany, 2022[29]	-	Mercedes-Benz (L3 ALKS, Dec 2021)[30]
UK	✓	Spring 2022[31], [32]	-	-	-	-
China	✓	General legislation, Apr 2021[33] Classification, Mar 2022[34]	-	Conditional	-	Baidu, Pony.ai (Beijing, Nov 2021)[35]
Japan	✓	May 2020[36]	Early 2022[37]	Driver-less with remote monitor[38]	-	Honda (L3 for congestion, Nov 2020)[4]
Korea	✓	Jan 2020 [39], [40]	-	-	-	-

* As of April 2, 2022, 32 of the 50 states in the US allow AV testing [41] ** As of April 2, 2022, 18 of the 50 states in the US allow deployment of AV, but to the authors' knowledge, details for the level of autonomy classification and due responsibility are not explicitly specified [41] *** As of April 2 2022, 20 of the 32 states that allow AV testing do not have requirements of safety driver if certain conditions are met [41] **** Cruise applied for driver-less deployment in California in May 2021 [42], received test permit in June 2021 and partial deployment approval in Sep 2021 [43], but only received "drivered deployment" approval by California Public Utilities Commission in Feb 2022 [44].

and formal methods for safety verification; (3) we conduct a state-of-the-art literature review on formal methods for guaranteeing safe automated vehicle motion control. Lastly, we point out trends and opportunities in the field of formal safety verification for ADS.

This survey is structured in the following way: section II summarizes the progress and challenges in AV safety certification. Section III overviews safety goals and state-of-the-art verification and validation approaches. Section IV provides the preliminaries for formal safety verification. Section V introduces a unified scenario coverage framework for both formal and sample-based methods. Section VI summarizes state-of-the-art formal control safety verification methods, whereas section VII summarizes state-of-the-art control synthesis approaches using formal methods. Section VIII discusses the open questions in formal safety verification and validation and lists resources available online. Section IX concludes the findings of this survey.

II. CERTIFICATION

A. Certification Status Worldwide

Since the piloting of automated vehicle programs such as the PROMETHEUS project [45] (1987-1995) and the DARPA Grand Challenge [46] (2004-2007), the development of AV has entered the era of on-road deployment. An incomplete summary of the current certification status of automated driving on public roads is provided in Table I.

B. Safety Certification Challenges

Among all the concerning issues, four particular challenges stand out for AV safety certification: the determination of unavoidable collision, the determination of liability, the verification cost for reasonable scenario coverage, and the additional cost of re-verification for ADS update.

Determining whether a collision situation is avoidable is not just an academic problem but also a liability-related legal problem. Despite the high expectations for autonomy, as humans, we can only expect robotic systems to optimize actions for the best possible outcome. For example, one can only expect automated

vehicles to apply evasive maneuvers to avoid an avoidable collision, but not an unavoidable collision. The requirement for such avoidability determination was passed as an amendment to the law in Germany in August 2021 regarding level 3 to level 4 function in HAV [28]. Specifically, in § 1e paragraph (2), number 2 of the law (Straßenverkehrsgesetzes, or Road Traffic Act) states that (English translation):

Motor vehicles equipped with an automated driving function must have technical equipment that is capable to independently comply with the traffic regulations directed at driving the vehicle and have an accident prevention system that:

- is designed to avoid and reduce damage,
- in the event of unavoidable alternative damage to different legal interests, the protection of human life being the highest priority, and
- in the event of an unavoidable alternative risk to human life, none provides further weighting based on personal characteristics.

In the legal text above, judging "unavoidable alternative damage" requires a formal determination of damage avoidability. Despite the different implementations of accident prevention systems in the industry, the avoidability judgment mechanism is better conservative than risky from a safety verification standpoint. At the same time, it is also undesirable to be "too conservative" to not use emergency maneuvers when an incoming collision is still avoidable. To illustrate this subtle difference, two situations are given:

- First, when the vehicle is very close to an unavoidable collision situation yet there is still room to avoid the collision, we do not want the ADS to make a false-positive unavoidable judgment and start preparing for the impact;
- Second, when the vehicle is actually in an unavoidable collision situation and not colliding yet, from a safety verification standpoint, the ADS should be sensitive enough to start preparing for the impact.

These two situations indicate that the accident prevention system in ADS should be able to make a clear-cut distinction between avoidable and unavoidable situations early enough. It should not hesitate to engage in a conservative, evasive,

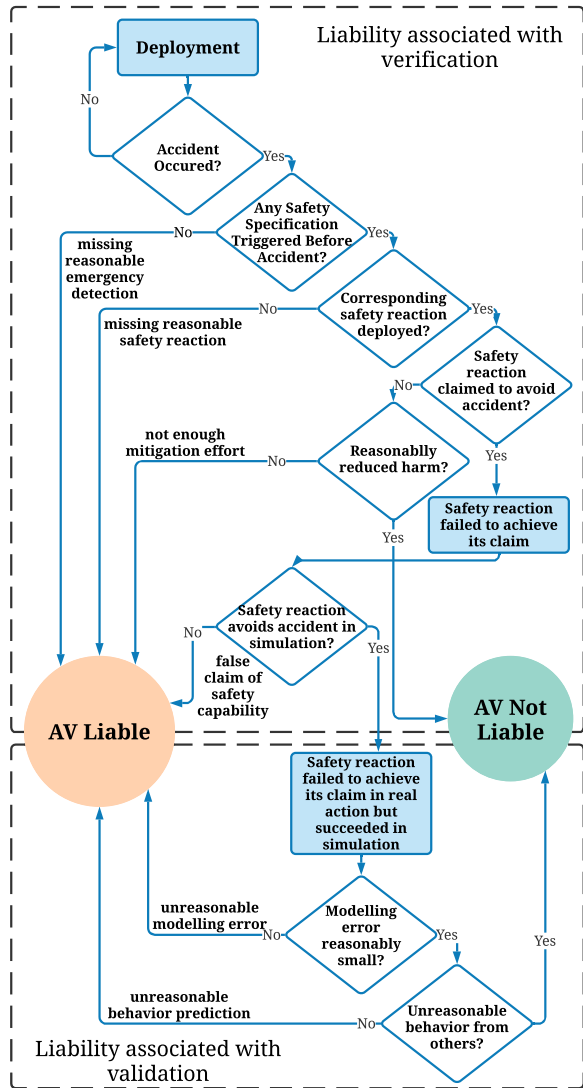


Fig. 1. Proposed liability determination framework using formal method, for Level 4 and Level 5 ADS.

or impact-preparation maneuver. Otherwise, according to the German law amendment [28], AV control is liable for collision damage.

Liability determination is also an emerging issue. At Level 3, the ADS is not liable as long as unexpected or difficult situations are handed over early enough to the human driver. The ADS is only expected to guarantee safety in its limited ODD environment. Even if the human driver does not take over in time, a fail-safe maneuver such as braking, stopping, and turning on hazard lights can free the ADS from liability in most cases. When a Level 4 or above AV is equipped with formal safety verified ADS, the liability determination can be streamlined according to a formal logic, illustrated in Fig. 1. For ADS developed with sample-based safety verification, it is yet to see how liability determination can be performed.

Another challenge for certification is the cost associated with verifying and validating the ADS. For an ADS industry stakeholder, already huge investments have been made to verify and

validate the in-house ADS. When it comes to type approval or certification, it is unclear how much of the V&V effort from the industry stakeholder can be “credit transferred” towards the safety certification.

Moreover, as more learning-based functions are incorporated into AV control design, the resulting control stack is constantly evolving based on collected learning data. The continuous learning and development of ADS software by the engineering team are adding another layer of uncertainty to safety V&V and certification. A question remains unanswered on whether regression testing is required for each ADS software update.

III. AUTOMATED VEHICLE SAFETY VERIFICATION AND VALIDATION

A. Qualitative Safety Goal

Safety is a system state free from harm. Furthermore, safety assurance is the capability to be free from risks developing into harm when they emerge [47]. Although huge safety enhancement potential exists for AVs as 94% of road accidents are driver-related [48], limitations of current safety V&V measures for AVs are widely recognized. As pointed out by the RAND Corporation report *Driving To Safety* [49], tens of billions of miles of on-road testing is required to ensure an acceptable level of AV safety for all possible scenarios and inputs. Whereas programmatically generated simulation tests, although performed much faster, more flexible in scenario generation [12], [50] and more economical, are susceptible to fidelity and realism concerns [9]. Even worse, both on-road testing and simulated testing cannot exhaustively cover all driving scenarios even for a small operational design domain (ODD) [23].

The widely used safety guideline standard *ISO 26262 Road Vehicles—Functional Safety* [51] is suited only for mitigating known unreasonable risks related to known component failure (i.e., known unsafe scenario). However it is not addressing AV driving risks due to complicated environment variants and how the ADS reacts to them, while nothing in the vehicle is technically failing.

Given the safety challenge above, a qualitative goal was proposed in *ISO 21448 Safety of the Intended Functions (SOTIF)* [52], which describes high-level goals of *minimizing known and unknown unsafe scenario outcomes* for ADS function design (Fig. 2). The challenge for meeting this goal, however, is the intractability of traditional methodologies such as field operational test (FOT) [13] to cover all possible scenarios for automated driving during testing.

Despite the challenge, there are promising methodologies and directions in safety analysis to move toward the ISO 21448 goal, such as maximizing scenario coverage in simulated testing [12], or creating safety guarantees through formal methods [20], [21].

B. Quantitative Safety Threshold

For AV testing, the large number of tested miles can be deceptive and does not always equal adequate scenario coverage. Specifically, if most of the driving are done on repeatedly tested, low-traffic highways, the “per-mile-value” of data quickly deteriorates. In fact, event-rates-per-exposure notions

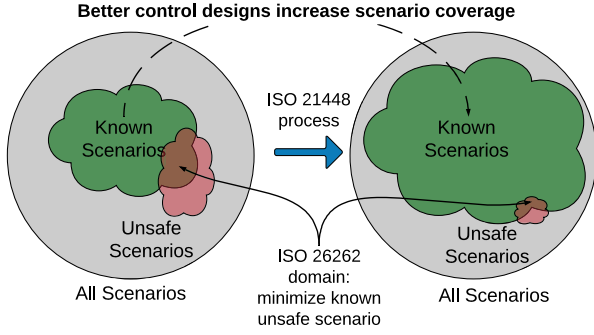


Fig. 2. The goal of ISO 21448 (SOTIF) [52]: minimize unsafe scenario outcomes, both known and unknown to the control design. Colors in the diagram represent: known safe scenarios (green), known unsafe scenarios (light red), unknown safe scenarios (grey), unknown unsafe scenarios (dark red).

such as injuries/fatalities per 100 million miles mentioned in the RAND report [49], or disengagements per 1000 miles used for California DMV disengagement report [53] are only rough criteria to provide an intuitive summary of how statistically safe is a particular operation, with a risk of selective bias.

In legal terms, *safe enough* means satisfying legally what *safety* is. Most US states demand that insurance is purchased to enforce the safety liability in AV testing, with an average amount of \$5 million US dollars [41]. Recently, vehicle manufacturers such as Mercedes-Benz [54] have accepted liability for accidents during the intended use of the ADS functions.

In technical terms, *safe enough* usually indicates full or sufficient scenario coverage for the designated ODD [13], [14]. In reality, relatively weak requirements have been seen in existing regulations [19] that only require checking “several scenarios that are critical”. A promising attempt to assign “volume” to scenario samples is seen in [22] by introducing a “ δ -neighborhood” around a discrete scenario. Mathematical algorithms such as T-wise [55], [56] and Poisson process [57] have also been adopted to achieve “almost full” statistical coverage by cleverly selecting candidates with limited samples.

Suppose that each single sampled scenario has a “coverage volume” (illustrated in shaded areas in Fig. 4, and represented by gridded units in Fig. 6(f)-(j)). Then the complete scenario coverage task for sample-based methods is the following: perform a sufficient amount of sampled verification tests to ensure that each “coverage volume” unit is either associated with one safely verified test result or proven to be accident-inevitable. When it comes to type approval and certification, the authorities need to set an acceptable success threshold to match up with public expectations. The simplest threshold r could be the ratio between the number of different tests verified safe N_{sv} and the number of minimum tests required for the ODD N_{ODD} :

$$r = \frac{N_{sv}}{N_{ODD}} \quad (1)$$

In section V we expand this scenario coverage concept to construct a unified framework of scenario coverage.

■ Sample-based Methods
 ■ Formal Methods
 ■ Field Operational Tests

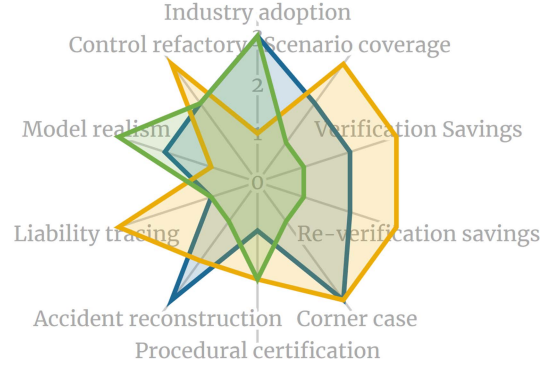


Fig. 3. Comparison of different verification and validation methods. **Industry adoption:** sample-based methods have been widely used to generate virtual scenarios to complement limited road tests, while formal methods are less common. **Scenario coverage:** in a properly abstracted model, formal methods can cover the entire scenario space by giving explicit specifications, while sample-based methods require large numbers of samples to establish coverage, and road tests can hardly cover corner cases. **Verification savings:** Based on proofs and dynamics, formal methods can save the most in verification cost, while road tests are more expensive than sample-based virtual tests due to hardware and personnel costs. **Re-verification savings:** formal methods save the most for re-verification since often only a minor update in the proof is needed; the need for regression testing after control update makes sample-based and especially road tests expensive. **Corner case:** corner case generation is well studied for sample-based methods, whereas formal methods do not explicitly focus on corner cases but cover them; for road tests, corner cases are difficult to encounter. **Procedural certification:** formal methods have the potential to fit with certification with its rigor in theoretical proof. At the same time road tests are practically implementable but coverage-wise insufficient, while sample-based testing lacks certifiable validity. **Accident reconstruction:** can be easily performed using sample-based scenario generation, but not economical to be reproduced with real vehicle tests; formal methods can utilize accident information to recheck the execution of formal safety specifications for design flaws. **Liability tracing:** can be performed naturally with formal methods as shown in Fig. 1. **Model realism:** is well matched in road tests, less matched in sampling-based simulation tests, and least matched in models used for formal methods due to the known issue of scalability. **Control refactory:** can be relatively easy with formal-methods-based verification, as long as new designs go through the same formal verification pipeline, whereas in sample-based testing or road tests, new extra efforts are unavoidable to re-verify the changes.

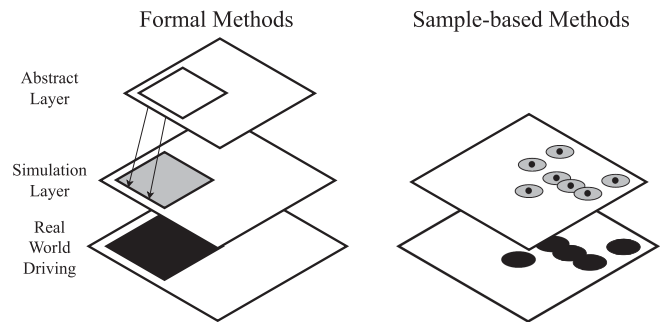


Fig. 4. Scenario coverage comparison between formal methods and sample-based methods: formal methods start from safety specification at a more abstract layer, may have a larger single coverage volume in scenario space, but the procedure to integrate formal specification into control synthesis or monitoring can be controller math dependent and tedious; sample-based methods have a more sporadic spread of scenario coverage due to the randomized generation process but starts covering cases directly at the simulation layer, making the sampling process straightforward to implement. Both approaches are trying to project maximum verified scenarios into real-world driving, but the difference between the simulation layer and real-world driving always exists.

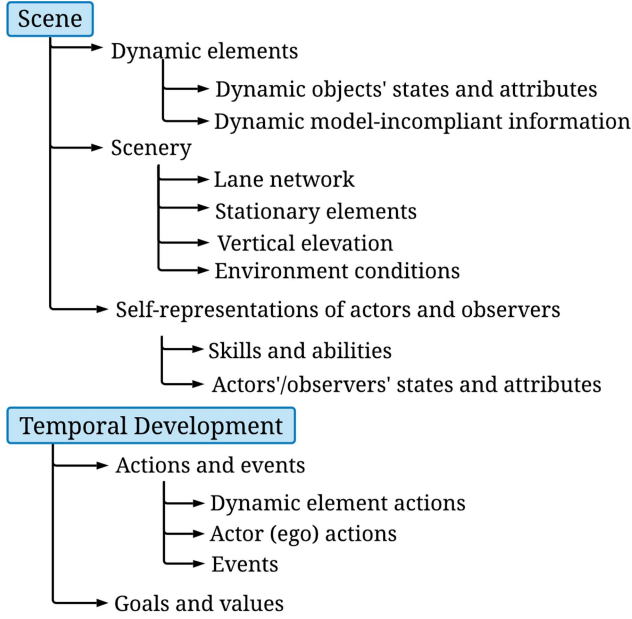


Fig. 5. Dimensions of scenario variation: according to ISO 21448 [52], the initial scene associated with a scenario can have multiple variations, including dynamics elements, scenery setting, and self-representations of actors and observers; given an initial scene, the temporal development can have additional variations of dynamic element actions, actor (ego) actions and goals and values of different participants in the scenario.

C. State-of-the-Art Verification and Validation Methods

The verification and validation methods can be categorized into three main groups: sample-based methods, formal methods, and field operational tests (FOT).

Sample-based methods verify AV control strategies by testing the strategies against diversified scenario samples. This is done either by maximizing scenario coverage with minimal scenario generation effort or finding attention-worthy safety corner cases such as high safety *criticality* scenarios [58]. Readers can refer to dedicated surveys [12], [14] for detailed discussion.

Compared to sample-based methods, formal methods have the advantage of high statement reliability [14] due to their rigorous logical foundation. Commonly used formal methods in AV safety include model-checking, reachability analysis, and theorem proving [15]. Model-checking originates from software development to ensure that the software behavior is according to the design specifications. When safety specification is expressed in axioms and lemmas as in [59], then theorem proving can be used to verify safety using worst-case assumptions. Reachability analysis [60] has a special place among the three due to its inherent capability of producing safety statements for dynamical systems [61], which capture the main characteristic of dynamic driving task (DDT).

Real world road testing, or field operational testing (FOT), is the ultimate yet expensive AV verification and validation approach. In a certain sense, it is the only way of true validation. Nevertheless, the shortcoming of FOT is also obvious: it cannot reach enough scenario coverage (especially near collision and collision scenarios) with a candidate AV controller on board.

A comparison of sample-based, formal, and field operational testing methods is illustrated in Fig. 3. The different aspects are scored on a scale of 0 to 3, with 3 representing highest satisfaction. In practice, often a combination of methodologies [15], [62] is necessary for the safety functions development cycle. One such novel approach is to combine sampling with resolution or probabilistic completeness with robust controlled invariant sets [22].

IV. FORMAL SAFETY VERIFICATION

Formal methods are a class of methods that applies mathematically rigorous techniques (usually in the form of logic calculi) to achieve specification and verification of software and engineering design [63]. In this section, we walk through the two essential concepts in formal safety verification, i.e., reachability analysis and signal temporal logic.

A. Reachability Analysis

We first define a dynamical system with continuous states and controls:

$$\dot{x} = f(x, u) \quad (2)$$

where state $x \in \mathcal{X} \subseteq \mathbb{R}^n$ and control $u \in \mathcal{U} \subseteq \mathbb{R}^m$ are measurable and continuous within their domains, $\mathcal{X}, \mathcal{U}$ are compact, $m, n \in \mathbb{R}$. The function (i.e. flow field) $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is assumed to be uniformly continuous, bounded and Lipschitz continuous.

Definition 1. (Maximal Forward Reachable Set): given a dynamical system (2), the maximal forward reachable set at time $t > t_0$ of a set of initial states $\mathcal{X}_{\text{init}}(t_0) \subseteq \mathcal{X}$ is:

$$\begin{aligned} \max FRS : \quad & \{x(t) \in \mathcal{X} \mid \exists u \in \mathcal{U}, \exists x(t_0) \in \mathcal{X}_{\text{init}}(t_0), \\ & x(t) = x(t_0) + \int_{t_0}^t f(u, t) dt\} \end{aligned} \quad (3)$$

Forward reachable set propagates dynamics to find all possible reachable states in a future time t since the current time t_0 from within $\mathcal{X}_{\text{init}}$. On the contrary, backward reachable set back-propagates to find all possible states from a previous time t that can lead to some $\mathcal{X}_{\text{goal}}$ at the current time t_0 :

Definition 2. (Maximal Backward Reachable Set): given a dynamical system (2), the backward reachable set at time $t < t_0$ of a set of target states $\mathcal{X}_{\text{goal}}(t_0) \subseteq \mathcal{X}$ is:

$$\begin{aligned} \max BRS : \quad & \{x(t) \in \mathcal{X} \mid \exists u \in \mathcal{U}, \exists x(t_0) \in \mathcal{X}_{\text{goal}}(t_0), \\ & x(t_0) = x(t) + \int_t^{t_0} f(u, t) dt\} \end{aligned} \quad (4)$$

The operator $\exists$ before u in the above expressions can be replaced with $\forall$ to create the “minimal” reachable sets, for example, minimal backward reachable set is:

$$\begin{aligned} \min BRS : \quad & \{x(t) \in \mathcal{X} \mid \forall u \in \mathcal{U}, \exists x(t_0) \in \mathcal{X}_{\text{goal}}(t_0), \\ & x(t_0) = x(t) + \int_t^{t_0} f(u, t) dt\} \end{aligned} \quad (5)$$

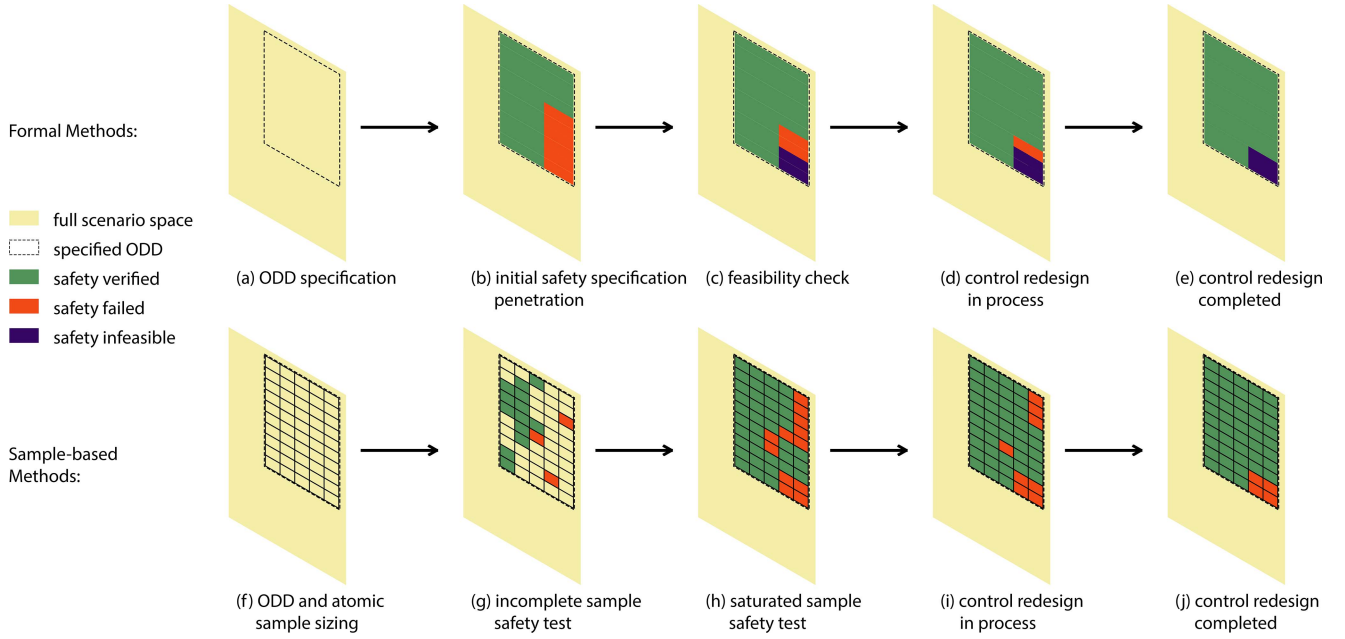


Fig. 6. Control policy evolution for safety verification. In (a)-(e), formal methods start with safety specification for the ODD (a), then an initial safety specification penetration test is performed to see how good the safety specification is kept (b). Then the feasibility check is performed to see how many failed scenarios are actually safety feasible (c). Then the AV controller is redesigned to remedy the failed scenario volume region(d), and eventually, all safety feasible scenario volume is verified safe with the candidate AV controller. In (f)-(j), the sample-based methods start by segmenting the ODD into verifiable scenario units (f), then an incomplete sampling safety test is performed to check for significant issues with the candidate AV controller (g). Subsequently, fully saturated sampling is performed to ensure scenario coverage (h). Controller weakness is exposed after a full scenario coverage test, and the redesign process iterates (i). Eventually, the controller's weakness stops shrinking, and the control redesign process is likely finished(j). Notice that safety infeasible scenario regions can be identified with formal methods, and no further redesign is needed once all volume in the ODD except for the safety infeasible scenario volume is verified.

In some cases, reachability within a defined time horizon is of more interest. Thus the definitions in the previous section can be extended to include time from the current time to the end of the time horizon. For brevity, only the definition of the maximal forward reachable tube is provided; the other variants can be easily derived accordingly:

Definition 3. (Maximal Forward Reachable Tube): given a dynamical system (2), the forward reachable tube during time $[t_0, t]$ of a set of initial states $\mathcal{X}_{\text{init}}(t_0) \subseteq \mathcal{X}$ is defined as:

$$\text{maxFRT} :$$

$$\{x(s) \in \mathcal{X} \mid \exists u \in \mathcal{U}, \exists s \in [t_0, t], \exists x(t_0) \in \mathcal{X}_{\text{init}}(t_0),$$

$$x(s) = x(t_0) + \int_{t_0}^s f(u, t) dt\} \quad (6)$$

In situations where the interaction among different traffic participants plays a critical role, it becomes imperative to include the influence of participating, sometimes even adversarial agents, into the reachability analysis. In this case, an additional input d is introduced to the system (2) to represent such adversarial control input:

$$\dot{x} = f_a(x, u, d) \quad (7)$$

where $d \in \mathcal{D} \subseteq \mathbb{R}^p$, $p \in \mathbb{R}$ is the adversarial input to the system. As an example, it can be control decisions of other vehicles where $f_a : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}^n$ represents the joint or relative dynamics of all participating vehicles. In such systems, the influence of adversarial inputs is not up to the ego vehicle control and, therefore, must be modelled conservatively to provide

guarantees in the worst cases. The definition of maxFRS in such systems is provided below as an example.

Definition 4. (Maximal Forward Reachable Set under Adversarial Influence): given a dynamical system as in (7), the forward reachable set at time $t > t_0$ of a set of initial states $\mathcal{X}_{\text{init}}(t_0) \subseteq \mathcal{X}$ is defined as:

$$\{x(t) \in \mathcal{X} \mid \exists u \in \mathcal{U}, \forall d \in \mathcal{D}, \exists x(t_0) \in \mathcal{X}_{\text{init}}(t_0),$$

$$x(t) = x(t_0) + \int_{t_0}^t f_a(x, u, d) dt\} \quad (8)$$

While the above definitions have been made in a deterministic sense, reachability can be stochastically interpreted, e.g., by determining the probability of system states reaching a particular target set [64].

B. Signal Temporal Logic

In the field of computer science and robotics, Signal temporal logic (STL) is a common language versatile for expressing and specifying requirements that are time-critical and contain continuous variables. The basic semantic of signal temporal logic is listed in table II, adapted from [20]. In brief, STL uses first order logic¹ to make statements about variables in their temporal development.

¹First-order logic: a collection of formal systems that uses quantifiers over non-logic objects such as variables. For example, while propositional logic can only make statements like "the vehicle violates lane boundaries," first-order logic can incorporate variables and state "there exists a coordinate C such that C is inside the vehicle occupancy area and C is outside the intended lane boundaries

TABLE II
BASIC STL SEMANTICS

STL semantic*	Equivalent STL semantic	Explanation
$(s, t) \models f(\cdot) \geq 0$	$f(s(t)) \geq 0$	$f(s(t)) \geq 0$ holds true for the trajectory (s, t) at time t
$(s, t) \models \neg(\phi)$	$\neg((s, t) \models \phi)$	The negation of predicate ² ϕ holds for the trajectory (s, t) at time t
$(s, t) \models \phi \wedge \psi$	$(s, t) \models \phi \wedge (s, t) \models \psi$	Both predicates ϕ and ψ hold simultaneously for trajectory (s, t) at time t
$(s, t) \models \phi \vee \psi$	$(s, t) \models \phi \vee (s, t) \models \psi$	Either predicate ϕ or ψ holds for trajectory (s, t) at time t
$(s, t) \models \Box_I \phi$	$\forall t' \in I \oplus t \quad s.t. \quad (s, t') \models \phi$	Predicate ϕ holds for the entire interval I
$(s, t) \models \Diamond_I \phi$	$\exists t' \in I \oplus t \quad s.t. \quad (s, t') \models \phi$	Predicate ϕ holds at least once in the interval I
$(s, t) \models \phi \mathcal{U}_I \psi$	$\exists t' \in I \oplus t \quad s.t. \quad ((s, t') \models \phi \wedge \Box_{[0, t']} \phi) \wedge ((s, t') \models \psi)$	Predicate ϕ always holds until predicate ψ is true at some moment t' in interval I . After moment t' predicate ϕ or ψ no longer has to hold

*Note: when I is dropped in the subscript (as is most cases), it indicates that $I = [0, \infty)$ by default.

As an example, we start from a casual statement in natural language, “(the AV shall) never cause a collision in traffic”. To reshape this statement into STL, some of the ambiguities need to be removed first. The word “cause” alludes liability assignment, which is not defined in STL quantifiers. So we replace “cause” with the quantifier “be in,” which is liability-neutral. Then the casual specification becomes “never be in a collision in traffic”. Second, STL requires a precisely defined scope of time, and for automated driving, it is a common practice to use the concept of the time horizon. Therefore the casual specification can be further modified into “never be in a collision in traffic during the time horizon.” In STL formula this is represented as:

$$(s, t) \models \Box_I \phi^{\text{collision-free}} \quad (9)$$

This expression means that during the time horizon interval I the AV trajectory always satisfies the requirement $\phi^{\text{collision-free}}$ (the AV state is in the collision-free set $\mathcal{S}^{\text{collision-free}}$).

The above STL-translated safety specification still stays at an abstract level in the set language, and it is the responsibility of AV control algorithm designers to honor this safety specification, by either synthesizing it into the controller architecture, or by performing safety verification over the designed controller to ensure that the specification is met with enough confidence or solid proof. Some examples of STL-based safety specifications are: (1) always keep the ability to return to the right-most lane within 5 seconds; (2) maintain collision-free and within road boundaries. When a finite time horizon (such as $I = [0, 25](s)$) is provided, the specifications (1) and (2) can be expressed as [20]:

$$(s, t) \models \Box_{[0, 25]} \Diamond_{[0, 5]} \psi^{\text{lane}} \wedge \Box_{[0, 25]} \phi \quad (10)$$

where ψ^{lane} represents the specification of being always within 5-seconds of returning to the right lane, and ϕ represents collision-free and within road boundaries. The expression $\Box_{[0, 25]} \Diamond_{[0, 5]} \psi^{\text{lane}}$ encodes “always” ($\Box_{[0, 25]}$) “have the existing capability to satisfy in 5 seconds” ($\Diamond_{[0, 5]}$) the “going back to the right lane” requirement (ψ^{lane}). The expression $\Box_{[0, 25]} \phi$ encodes “always” ($\Box_{[0, 25]}$) “satisfy collision-free and road boundary constraints” (ϕ). A more sophisticated control goal can be encoded based on this STL expression, for example, to indicate the task of overtaking a slow car in front (expression

modified from [20]):

$$(s, t) \models \Box_{[0, 25]} \Diamond_{[0, 5]} \psi^{\text{lane}} \wedge (\phi \mathcal{U}_{[0, 25]} \psi^{\text{pass}} \vee \phi \mathcal{U}_{[0, 25]} \psi^{\text{stay}}) \quad (11)$$

where ψ^{pass} represents successful overtaking of the slow front car, and ψ^{stay} represents staying behind the slow front car. The expression $(\phi \mathcal{U}_{[0, 25]} \psi^{\text{pass}} \vee \phi \mathcal{U}_{[0, 25]} \psi^{\text{stay}})$ specifies two alternative goals (ψ^{pass} or ψ^{stay}) on top of the safety constraint ϕ . STL expressions like equation (11) were implemented into control synthesis via a connection between STL and reachability analysis in some recent work [20]. Apart from control synthesis, safety specifications expressed in STL can be used as assertions³ during controller prototyping to indicate safety violations, so that developers are constantly aware of safety specifications during control design. The logic calculation for determining the truth of a safety specification is often done by solving a satisfiability modulo theory (SMT) problem⁴.

C. The Formal Safety Philosophy

In the formal safety philosophy, a safety specification is either fulfilled or violated, and the fulfillment property can be **designed** by integrating the specification satisfaction into the model-based control synthesis. After such design is complete, the burden of ensuring such specification fulfillment is transferred to online validation of model correctness: the system is provably safe as long as the control-oriented model is validated to be correct. However formal methods suffer from poor scalability [9]. High system complexity with continuous dynamics has been a major limiting factor for its wide application in automated driving.

V. UNIFIED SCENARIO COVERAGE FRAMEWORK

A. Scenario Space and Scenario Volume

According to ISO 21448 [52], *scenario* is a description of the temporal development between several scenes in a sequence of scenes, and *scene* is a snapshot of the environment, including the scenery, dynamic elements, and all actor and observer self-representations, as well as the relationship among those entities. According to this definition, the task of *ensuring full safety* in a defined ODD can be described as performing safety verification

³Assertion: In computer science, an assertion is a predicate connected to a point in the program that should always evaluate to true if the program is implemented correctly.

⁴Satisfiability modulo theory (SMT) problem: true or false determination for formulae expressed using first-order logic. for reference see [65].

B.” In the latter statement, “exists” and “inside” are quantifiables, and C is a variable.

and validation for all possible scenario variations within the ODD. The scenario variations within an ODD are two-fold: variations of the initial scene, and variations of the temporal development since the initial scene. The different dimensions of scenario variation are illustrated in Fig. 5.

Given a specific ODD designated as $\mathcal{O}$, one can define an entity to span all possible scenarios within the ODD description. We call this entity the *scenario space*. Using parameterization, the *scenario space* is defined as:

Definition 5. (Scenario space) The N -dimensional subspace $S(\mathcal{O}) \subset \mathbb{R}^N$ where $S(\mathcal{O}) = \{(p_1, \dots, p_N, q_1, \dots, q_M) \in \mathbb{R}^{N+M} | p_i \in [\underline{p}_i, \overline{p}_i], q_j \in \mathcal{Q}_j\}$ is the scenario space for an ODD $\mathcal{O}$ specified by the continuous scenario parameters $p_i (i = 1, \dots, N)$ and the discrete scenario parameters $q_j (j = 1, \dots, M)$. $\underline{p}_i, \overline{p}_i \in \mathbb{R}$ are the lower and upper bounds of each continuous parameter p_i , $\mathcal{Q}_j \subset \mathbb{R}$ is the set of possible values for each discrete parameter $q_j \in \mathcal{Q}_j$.

Remark 1: Note that no description of stochastic behavior is explicitly given in such a parameterized definition since the parameters are used to capture the scenario event as if it has already happened and is fully describable without stochasticity.

Assume that the number of continuous dimensions of the scenario space $S(\mathcal{O})$ is N , the number of discrete parameter combinations is $\prod_{j=1, \dots, M} |\mathcal{Q}_j|$, where $|\mathcal{Q}_j|$ is the cardinal of parameter set $\mathcal{Q}_j$. Assume the feasible range of each continuous parameter p_i is $[\underline{p}_i, \overline{p}_i]$, then the total *scenario space volume* can be defined as:

Definition 6: (Scenario space volume): The scalar value $V_{S(\mathcal{O})} = \prod_{i=1, \dots, N} (\overline{p}_i - \underline{p}_i) \cdot \prod_{j=1, \dots, M} |\mathcal{Q}_j|$ is the scenario space volume of the scenario space $S(\mathcal{O})$.

Full scenario coverage for requires that the verified scenario space volume V_v encompasses the volume $V_{S(\mathcal{O})}$:

$$V_v \supseteq V_{S(\mathcal{O})} \quad (12)$$

where V_v is the safety verified scenario volume.

Similarly, *partial scenario coverage* indicates that safety verification is carried out only on a portion V_v' of the scenario space volume $V_{S(\mathcal{O})}$:

$$(V_v' \cap V_{S(\mathcal{O})}) \subset V_{S(\mathcal{O})} \quad (13)$$

The scenario coverage ratio is therefore:

$$\frac{V_v' \cap V_{S(\mathcal{O})}}{V_{S(\mathcal{O})}} \quad (14)$$

B. Unified Scenario Coverage

Sample-based methods do not explicitly possess the volumetric property associated with each sample scenario, as each scenario sample is rather atomic and volume-less. In order to make a unified comparison and utilization of both methods for safety verification, notions of scenario volume must be assigned to sample-based methods. For simplicity, assume all N continuous dimensions are orthogonal to each other, and an axis-aligning polygonal volume in N -dimensional scenario space is assigned for a scenario sample specified by the parameter tuple $(p_1^0, p_2^0, \dots, p_N^0)$. The assigned volume will take the width of w_i in the i^{th} continuous scenario space dimension around the

parameter value p_i^0 , where w_i is the tolerable resolution provided by the verification requirement:

Definition 7. (Unit scenario volume): Given a scenario space with N continuous dimensions and tolerable resolution of $w_i, (i = 1, \dots, N)$ in each dimension, the unit scenario volume is:

$$V_0 = \prod_{i=1, \dots, N} w_i \quad (15)$$

Thus verifying full scenario coverage with sample-based methods requires generating distinct enough, evenly spaced samples such that the overlapped volume of all verified samples supersedes $V_{S(\mathcal{O})}$. The number of samples required is:

$$n = \lceil \frac{V_{S(\mathcal{O})}}{V_0} \rceil \quad (16)$$

where $\lceil \cdot \rceil$ is the ceiling function.

Remark 2: n can possibly be reduced if evidence can be provided to show that verifying specific scenario samples can simultaneously verify of a particular group of scenario samples.

We also define safety for this unified framework. Without loss of generality, *safety* can be defined as a specification in natural language:

Definition 8. (Safety Specification): Safety is the verifiable truthfulness of a statement specified by one or more safety requirement clause(s) $\mathcal{N}$ in natural human language.

As an example, $\mathcal{N}$ could be “the automated vehicle should be collision-free at all times” or “the automated vehicle should be fault-free at all times.” If translated to STL, these two safety specifications can be expressed as:

$$(s, t) \models \Box \phi^{\text{collision-free}}, \quad (s, t) \models \Box \phi^{\text{fault-free}} \quad (17)$$

In sample-based methods, the safety of a simulated scenario is checked at the end of each scenario sample. On the other hand, formal methods usually start with the safety specification before any simulation is performed. The specification is translated into a machine-interpretable language statement, for example, in linear temporal logic (LTL) or signal temporal logic (STL). Then the translated specification can be utilized by checking the validity of the specification in simulated/real-world tests or by turning the specification into system constraints or other control design functions during control synthesis.

Definition 9. (Specification translation): is the process of turning a specification $\mathcal{N}$ in Definition 8 to an equivalent verifiable expression $E_{\mathcal{N}}$ expressed in the verification model states (s, t) .

For example, the expression “the automated vehicle should be collision-free at all times” can be translated into a verifiable expression in the form of a mathematical function that evaluates occupancy overlap between the automated vehicle and any other objects in the simulation model and returns true if overlap happens in the simulation. When a control policy is specified, we have the following variant:

Definition 10. (Policy-controlled specification translation): is the specification translation under the influence of a control policy π to the system.

However, practical challenges prevent the ideal 100% translation of formal safety specifications due to the following reasons: first, formal methods rely on arguments based on an abstraction of the real world; therefore, discrepancies – whether large or small – between abstraction and reality will occur and impair the effective safety guarantee. Second, the actually developed controller usually has performance limitations that prevent compliance with the safety specification in every case. Third, practical scalability difficulties exist in translating formal safety specifications into safety verification or control synthesis. In other words, formal methods usually have a *penetration rate* less than 100% in reality. Formally we define the *penetration rate* as the percentage of verifiable scenarios in the scenario subspace specified by a formal safety specification:

Definition 11. (Specification penetration rate): The ratio between the specification-holding scenario volume V_h and the specification-claimed scenario space volume $V_{S(\mathcal{N}, \pi)}$ during a policy π -controlled specification translation is the specification penetration rate $\mathcal{PR}(V_{S(\mathcal{N}, \pi)}, \pi)$:

$$\mathcal{PR}(V_{S(\mathcal{N}, \pi)}, \pi) = \frac{V_h}{V_{S(\mathcal{N}, \pi)}} \quad (18)$$

Remark 3: Definition 11 allows us to quantify the performance of a specific control policy π in faithfully carrying out safety specification $\mathcal{N}$. Imperfections of any control policy are expected, but it is the responsibility of safety verification to quantify and curb such imperfections.

Practically, finding out about the specification penetration rate of a candidate controller for AVs in a specified ODD can be a challenge. First, deductive formal methods such as theorem proving have been limited to relatively simple discrete dynamic systems. Extending theorem proving to systems with continuous dynamics requires a more established framework (a promising example [66]) and a considerably higher computation resource. Second, traditional algorithmic formal methods such as model-checking [67] also do not apply to systems with continuous dynamics. Running exhaustive tests of all possible actions would also create a computation resource problem when fine discretization of continuous action space is needed. However, Recent progress in combining STL with reachability analysis [20] can be a possible solution to evaluate such penetration rate by calculating verification arithmetic on the candidate controller over the scenario space of the ODD, and predicting the volume portion where the candidate controller will fail the STL specification. The specification penetration rate calculation problem is still an open one, and future work may combine different ways of evidence gathering, both deductive and algorithmic, to offer different alternatives.

Definition 12. (Scenario safe coverage): The scenario safe coverage $\mathcal{C}_\pi(\mathcal{O})$ of a policy π to a scenario space in a specified ODD $\mathcal{O}$ is defined as the ratio between the safety verified volume and the full scenario space volume with the candidate policy π , either using the sample-based or the formal verification method.

For simplicity, in the case of only one safety specification $\mathcal{N}$, the corresponding scenario coverage expressed in the

sample-based and the formal methods are, respectively:

$$\mathcal{C}_\pi(\mathcal{O}) = \frac{V_v' \cap V_{S(\mathcal{O})}}{V_{S(\mathcal{O})}}, \quad \mathcal{C}_\pi(\mathcal{O}) = \frac{V_h \cap V_{S(\mathcal{O})}}{V_{S(\mathcal{O})}} \quad (19)$$

given that the specification-claimed scenario space volume $V_{S(\mathcal{N}, \pi)}$ supersedes the scenario space volume $V_{S(\mathcal{O})}$.

Remark 4: In terms of scenario safe coverage, both sample-based and formal methods can provide confidence based on mathematical rigor at the simulation layer. Definition 6 provides a unified framework for both schools of methods to have a level playing ground for comparison.

The route to complete scenario coverage is illustrated in Fig. 6. Both formal and sample-based methods are helpful tools to assist in discovering AV safety control weaknesses, and AV control developers can select either or both routes to their convenience and preference.

In summary, quantifying scenario coverage $\mathcal{C}_\pi(\mathcal{O})$ of a candidate AV control policy π in a specified ODD $\mathcal{O}$ (similar to solving the safety quantification problem in [22]) can be performed by either sample-based or formal methods, or even a mixture of both.

An illustrative example of the process for verifying scenario coverage in both formal and sample-based methods is provided in Fig. 6. A difference between the two methods is that formal methods have the inherent capability of determining a part of the scenario space where safety is infeasible through, for example, reachability analysis.

VI. STATE-OF-THE-ART FORMAL CONTROL SAFETY VERIFICATION

Safety verification can be performed either during simulation (online) with a prediction module or after simulation (offline) to check results (Fig. 7). When the control policy of ego and participating traffic agents are known (*white-box control*) or partially known (*grey-box control*), formal methods can leverage such information to narrow down the reachable sets to provide a ‘tighter’ prediction of vehicle motion, leading to valuable applications such as online safety monitoring and verification. On the other hand, if the control policy is proprietary and completely unknown (*black-box control*), the safety verification process will have to involve statistical learning of the black-box controller’s behavior after each simulation, which in turn will guide the selection of next scenarios to be tested to better target counter-examples.

A. Formal Safety Verification of Known Control Policy

If the control policy is fully known (usually only limited to ego vehicle), then safety verification can be performed with much certainty, effectively removing the variable u in the RA. This is usually done by computing reachable sets of controlled ego vehicle and comparing them with the occupancy set of obstacles in the interested time interval [68]. However, the value of verifying known control policy against a particular scenario is limited to observing the short-term effect of variables other than the vehicle control itself, such as disturbances.

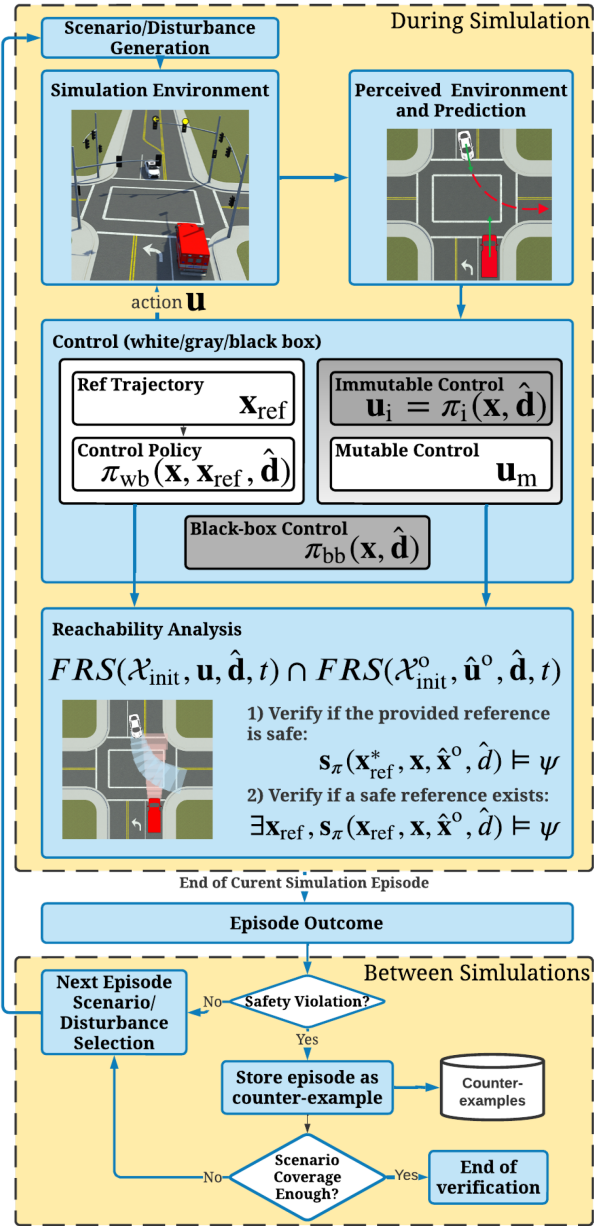


Fig. 7. The procedure of different verification schemes: A priori verification can be performed for white-box and gray-box controls, shown in the top yellow box. The black-box control policy can only be verified through a posteriori verification, shown in the yellow box at the bottom. The difference is that a posteriori verification can only be performed between simulations using the final safety outcome of each episode. In contrast, a priori verification can be performed as long as the perception of the environment is initiated in simulation.

B. Formal Safety Verification of Black-Box Control Policy

When the control policy is completely unknown, the verification needs to fall back onto more general plans that do not require control policy information. Such plans treat the controller-plant ensemble as an overall system, and investigate how the input (such as disturbance) to the overall system could lead to undesired output (such as safety violations). A survey on black-box verification [69] listed several such approaches as simulated annealing, evolutionary algorithms, Bayesian optimization, or extended ant colony optimization. However, due

to the black-box nature of the control policy, all conclusions of safety verification are probabilistic in nature.

C. Formal Safety Verification of Grey-Box Control Policy

If the control policy is partially known, modified forms of reachability analysis can be performed.

In the setting of a semi-autonomous vehicle where the human driver's behavior is estimated at best, the safety verification has to be performed with a partially known control policy (in this case, the human driver policy). A particular lane-keeping assist controller designed in this setting treats the human driver's steering policy as known and immutable while treating the human driver's acceleration policy as unknown and mutable [70]. Such treatment allows RA to predict the range of reachable states of the vehicle for all possible acceleration policies within a proper time horizon. The system can determine if the human driver needs assistance by comparing the current vehicle state with backward reachable safe sets.

When at least the qualitative objective of a control policy or function is known (e.g., activate intervention to save the vehicle from an avoidable collision due to driver mishandling), then RA can be used to verify if such control policy or function has stayed faithful to its qualitative description. For example, RA can be used to tell if a collision-avoidance system has missed or abused the opportunity to intervene [71]. Such grey-box control policy safety verification rely on problem-specific definition of what is partially known, and is therefore less scalable to more general driving tasks.

D. Combine Formal Verification With Prediction

To one extreme, if the worst-case assumptions are made for other vehicles' behaviors in a dense traffic setting, which is that the participating vehicles are all wanting to run into the ego vehicle, then the ego vehicle will likely have a collision every a few seconds. This is easily a too conservative and useless assumption. To another extreme, if other traffic participants are assumed to be all perfectly rule-obeying and acting like deterministic robots (e.g., driving at constant speed along with their lane centers), then the risk of underestimating participant behavior uncertainty is high and dangerous. Prediction errors can be minimized if all participating vehicles communicate under a vehicle-to-vehicle network [72], but this is often an ideal assumption.

The problem of behavior prediction can be a topic of its own right and often involves subtle distinctions of the information model and relates to game theory. For safety analysis, presumptions of behaviors can lead to interesting guarantees regarding accident liability. As a particularly interesting case [73], by modeling only traffic-obeying agents, it is claimed that legal safety can be guaranteed if ego vehicle trajectories are verified against such agents modeled as fully traffic-obeying. A similar assumption of "responsibility-sensitive safety" (RSS) is proposed in Mobileye's white paper for safety [74], assuming that other traffic participants should act according to some agreed "common sense," and the AV should plan accordingly. Efforts to bridge the gap between reachability-based safety and RSS are

TABLE III
FORMS OF REACHABILITY ANALYSIS DEPLOYMENT AND TYPES OF GUARANTEES PROVIDED

Reference	Form of Reachability Analysis	Type of Guarantee
[78], [79]	maxFRS + minBRS	guarantees collision if no solution for “anticipated reachable set” is found
[80]	FRS of controlled dynamics	predicts unavoidable collision within control policy
[81]	overapproximated maxFRS, maxFRT	all possible forward motions are included
[82]	inaccuracy around trajectory-planned vehicle	checks probability of collision within the inaccuracy model bounds
[83]	minBRT	guarantees collision free in horizon
[70]	maxBRS for partially-controlled & uncontrolled vehicle	conservative collision-free guarantee for non-extreme driving
[84]	overapproximated maxFRI	guarantee “not-at-fault” safety by checking potential collision in time intervals leading to end of time horizon
[85]	maxBRT	guarantees safe interoperability: possibly safe human takeover from autonomous driving system
[86]	maxFRS of uncontrolled dynamics	guarantees collision free in horizon
[87]	funnel (a variant of FRT for controlled vehicle)	guarantees collision-free if a funnel can be found to stay clear of obstacles at all times
[88], [89]	maxFRS implemented by Flow*[90]	guarantees safety of deep neural network (DNN) controlled close-loop system
[91]	maxFRS of (possibly) occluded vehicle	guarantees collision-free with possibly occluded vehicle(s)
[92]	maxFRS of each agent communicated through a decentralized network	real-time collision-free guarantee for a group of autonomous agents
[93]	maxFRS + collision state pruning + maxBRS	guarantees collision-free and successful reach of target state if “controller contract” (state constraint tubes) are honored
[94]	maxFRT + pedestrian intent prediction	guarantees safety within a high probability bound of pedestrian motion

seen in [75]. Different subjective beliefs of “aggressiveness” of other vehicles affect the calculated safe reachable sets [76]. As a general rule of thumb, more confidence in predicting human driving behavior can lead to less conservative AV planning and control [77]. It should be noted, however, that any such prediction is probabilistic by nature, and probabilistic safety verification should not be biased towards the safety of a particular group of road users.

VII. STATE-OF-THE-ART CONTROL SYNTHESIS WITH FORMAL SAFETY GUARANTEE

Formal safety guarantees have been developed for automated driving simulations and tests in the literature, mainly by performing reachability analysis. The types of guarantees and associated references are summarized in Table III.

A. Formal Control Mode Arbitration

Formal methods can provide valuable information on the criticality of the current driving situation. This information can be used to develop criteria for whether a different control scheme is needed to bring the ego vehicle to safety from a safety-critical scenario.

One such arbitration criterion is whether the vehicle is in an *inevitable collision state* (ICS) [95], which some argue is a more robust measure of collision proximity than traditional threat metrics such as time-to-collision (TTC) in estimating collision frequency [57], a vital quantity in determining automotive safety integrity level (ASIL) in ISO26262 [51]. Suppose the vehicle is indeed in such a state. In that case, an impact preparation should be executed (usually with extensive use of braking to slow down [96], or by following *not-at-fault* trajectories [84].) since collision is inevitable. This concept has been used for prototyping ADAS functions, such as determining when to trigger automatic emergency braking (AEB) [97], [98] or producing collision avoidance maneuvers [99]. Reachability analysis is

commonly used to determine such ICS. [100]–[103]. When the prediction of moving obstacles is crucial in dynamic traffic scenarios, a probabilistic framework should be adopted to create a probabilistic equivalent of ICS [104], [105]. For certain semi-autonomous or ADAS functions, the determination of whether the system is in an inevitable collision state can help determine whether ADAS intervention is missed or unnecessary [71].

Another arbitration criterion is whether target states are achievable under the current motion mode of the system. As an example, motion modes of a vehicle can be categorized using the notion of *trim trajectories* characterized by the constant input required to maintain the motion [106], [107]. A trajectory library of a specific system can be built for the controller to choose from [87], [108]. Under such a context, *continuous feasibility* and *control liveliness* are key concerns ensuring the accomplishment of higher-level tasks, such as overtaking another slow vehicle, merging into a busy fast lane, or avoiding an obstacle at high speeds [109]. When selecting a mode from a large library of complicated dynamic modes become computationally demanding, learning-based approaches such as reinforcement learning can be leveraged to efficiently learn the proper mode arbitration decision according to the environment [110], [111].

B. Formal Methods for Direct Control Integration

Reachability can also be directly integrated into the control synthesis “at all times,” providing continuous guidance/preference to the incumbent controller to avoid catastrophic events caused by infeasible control sequences.

One of such direct control integration methods is model predictive control (MPC), where formal methods, especially reachability analysis, can be used to develop robust MPC algorithms (Fig. 8). Reachability analysis is used to find the feasible set of states for each MPC planning horizon, so that the resulting MPC algorithm will not try to find the optimal solution in the non-feasible regions [112]–[115], leading to destabilization or control failure (Fig. 8). This is also called recursive feasibility [116]

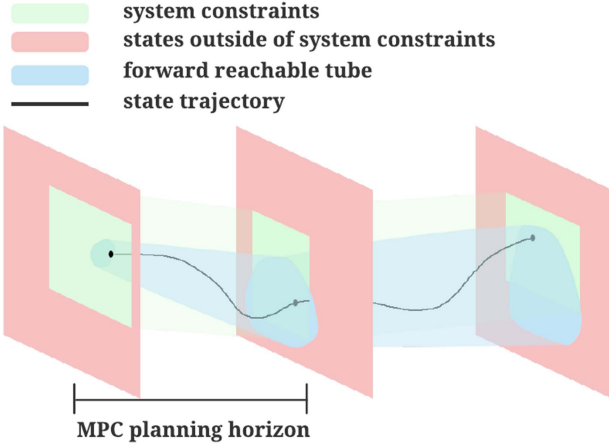


Fig. 8. Robust MPC can take advantage of reachability to calculate a feasible set of states at the planning horizon (i.e., terminal state constraints), thus ensuring that the optimization process will not end up solving an infeasible constrained problem.

or persistent feasibility [117]. A stochastic equivalent of robust MPC can be developed when the reachable set constraints are probabilistic [118]–[120]. Additional benefits of applying RA to MPC besides robustness and feasibility guarantee include: establishing system input-to-state stability [121], [122], separation of robustness and performance as in learning-based MPC [123], and reducing complexity in explicit MPC by removing regions that will never be reached [124].

Another way to integrate formal safety into control design is through safe reference trajectory generation (Fig. 9). Reachability can be used in the trajectory planning phase to generate optimally safe and physically-feasible trajectories for lower-level controllers to follow [86], [117], [125].

VIII. OPEN QUESTIONS AND RESOURCES

A. Open Questions

1) *Fidelity Speed Trade-Off*: The trade-off between dynamics fidelity and computation tractability in performing reachability analysis remains a significant limit. On many occasions, a complete reachable set (reachable states with any possible control inputs) is neither necessary nor economical. With a given control policy, the policy-bounded reachability can usually be computed faster as a subset of the complete reachable set. Moreover, the actual software/hardware implementation could further limit the configuration space. Considering such limits in the RA, it is shown [126] that up to 91% of computation load can be reduced.

In situations where actuation is less frequently demanded, such as cruising on the highway, the computation can be further reduced by intermittently performing reachability analysis and skipping computation while guaranteeing safety via control invariant set properties [127].

2) *Data-Driven Approach to Reachability*: An increasing number of studies are using data-driven approaches to address dynamical systems verification problems such as estimation of dynamical systems region of attraction [128] or reachable

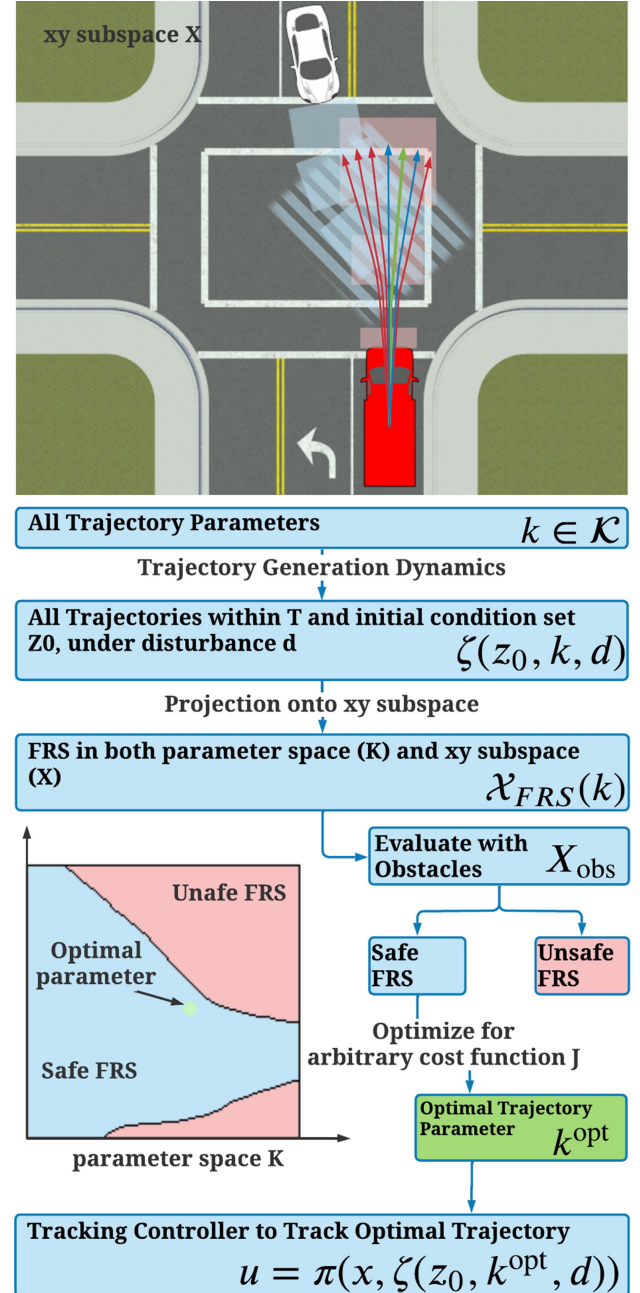


Fig. 9. Reachability-based safe trajectory design process: first possible trajectories are parameterized (arrows in the figure); then, the trajectory parameters are combined with initial states to produce FRS (red shaded area) and evaluated with obstacle occupancy (blue shaded area for the obstacle’s geometric center, blue striped area for occupancy at a particular time); the safe candidates are filtered, and a cost function selects an optimal set of trajectory parameters; finally a lower level tracking controller tracks the optimal trajectory.

set [129]–[132]. To some extent the close-loop DNN-controlled system can be verified for safety by transforming DNN into an equivalent hybrid system and by performing a reachability analysis on the obtained hybrid system [88], [89]. As data-trained controllers are increasingly being tested and applied in autonomous systems such as the aircraft collision avoidance system ACAS X [133], [134], the safety verification of neural-network-based controllers has become a research topic of

TABLE IV
NUMERICAL METHOD FOR REACHABLE SET/TUBE/INTERVAL CALCULATION

Method	Pros	Cons	References
Rectangle approximation	fast	accuracy suffers	[78], [79]
Polytope approximation	relatively fast, convex/nonconvex	usually only convex shapes have been used to exploit fast calculations	[155], [70], [135], [136]
Zonotope approximation	fast[157]	conservative[165] and convex	[81], [165]
Ellipsoid approximation	quadratic in nature, relatively fast, fits quadratic Lyapunov functions for calculation	shape may overly approximate vehicle occupancy	[166]
Star set approximation	shape more flexible than zonotope or polytope, non-convex, less over-approximation	less intuitive	[158]
Piece-wise linear system equation approximation	proved convergence, preserves the region of limit cycles	convergence speed can be improved	[159]
Semi-definite programming over polynomial representations of continuous functions	has efficiency improvement potential	slow in computation speed, currently not real-time implementable	[86]
Differential inclusion of nonlinear dynamic model	lower computation complexity	accuracy may suffer from dynamic over-approximation, but partition techniques[161] can partially make up this shortcoming	[160], [161]
Human-in-the-loop data sampling	fast since data sampling is offline, captures human driving pattern	data bias affects the validity of estimated set	[131]
Controllable set estimation using simulation-generated data	high data coverage thanks to simulation	relies on heuristic classification criteria for controllability	[129], [130]
Warm-start of HJI value function	guarantee over-approximation of true BRS, and reduces computation time compared to full initialization	the approximation to true BRS is not always exact, sometimes only conservative over-approximation can be achieved	[153], [152]
Local BRS update	significantly reduces computation time compared to full BRS update	limited to dealing with static obstacles in the environment	[152]
Koopman operator linearization of data-driven model	Fast reachable set computation, system can be a black box model	relatively new approach	[162]
Taylor model(TM) based flow-pipe tight over-approximation	tight over-approximation to nonlinear ODEs, fast calculation	n/a	[90]
Sample-based maxFRS under-approximation	the diversity of sampling is guaranteed by GreedyPack algorithm	GreedyPack is not the most efficient way to conduct diversified sampling	[163]
Sample-based discrepancy function over-approximation	works for black box dynamics model with white box transition graph	Subjects to learning bias	[167], [130]

interest. Feedforward neural network (FNN) based closed-loop controllers using ReLU activation functions can be verified at their output layers using exact or overapproximated reachable sets to comply with simple sets of linear state constraints [135]–[137]. While the collision modeling in aircraft systems involve higher dimensions and susceptible to aerodynamic uncertainties, the scenarios are actually more straightforward. It is yet to see how such neural-network-based collision avoidance mechanisms perform on automated ground vehicles. At the same time, some novel feedforward neural network training and verification using RA have been proposed for robotic systems [138].

While well-studied vehicle dynamics largely govern vehicle behavior, pedestrian behavior is much more unpredictable and relies more on data-based estimation [139]–[141], although some constraints such as velocity, acceleration, or jerk can be assumed to curb the reachability [142].

3) *Beyond Binary Safety Verification*: Instead of using reachability for binary safety verification (e.g., safe or unsafe), it can also assist decision-making in game-theoretic traffic setting to increase ego vehicle payoffs [143]. Although collision avoidance is a hard constraint that shall be met at all times to ensure safety, in a more general case, when different objectives of driving can be intermittently compromised, the study of formal verification of weakly-hard systems⁵ can become useful [144]. In fact, considering that human perception is flawed or even

delayed up to seconds [145], weakly-hard systems are not such a bad compromise.

4) *Building New Logic for Safety Description*: Although the majority of primary studies use linear temporal logic implicitly or explicitly for arguing safety guarantees, there exist developments of new forms of logic, such as *Differential Dynamic Logic*⁶ [146] (based on the concept in [147]), allows statements and related proofs such as “an earlier engagement of the emergency brake yields a lower collision speed”. It would be therefore interesting to see developments of new logic constructs for proving more sophisticated safety-related statements, as they could potentially help resolve liability issues generated by AVs.

5) *Ethical Verification With Reachability*: Ethical verification [148] is an emerging field in academia. Some edge cases exist where zero-casualty is infeasible, and decisions still have to be made based on ethical calculations [149]. Deontic logic [150], which argues about the logic of duty and responsibility, is experimented to fit into traffic rule and accountability calculations [151].

B. Resources

1) *Numerical Methods for Reachability Analysis*: Reachable sets originate from the evolution of the set of states in system dynamics. Therefore there are two major system-model-based

⁵Weakly-hard system: systems that allow certain degree of constraint or deadline misses. The degree of allowed misses can be characterized by the ratio of missed versus all constraints/deadlines.

⁶Differential dynamic logic(dL): a logic for specifying and verifying hybrid systems, verifiable with dL verification calculus, explained at <https://symbolaris.com/logic/dL.html>.

methods to approximate reachable sets numerically, i.e., set shape approximation [152], [153] and system dynamics approximation [81], [87], [88]. Apart from model based methods, data sampling is another alternative to approximate reachable sets [154] or their control invariant subsets [129], [130]. Set contour approximation [70], [78], [79], [135]–[137], [152], [153], [155]–[158], dynamics approximation [86], [90], [159]–[161], and sample-based approximation [129]–[131], [162], [163] are viable numerical options. A summary of methods used in primary studies is listed in Table. IV. Academic competitions for reachability analysis exist, such as ARCHCOMP [164], which is a competition of scientific software in the context of algorithmic verification of continuous and hybrid systems.

2) *Software Tools for Reachability Analysis*: For the reader's convenience, a set of reachability calculation software tools have been identified: Level-set Toolbox, helperOC, BEACLS, FaSTrack, KeYmaera X, MPT, dReach, SpaceX, CommonRoad, SPOT, CORA, Flow*, ARIANDE, HyCreate, Hylaa, NNV, JuliaReach, C2E2, DryVR, AROC⁷. Apart from various calculation approaches, efforts in constructing benchmark for reachability tests [168] also help to compare different tools.

IX. CONCLUSION

This paper presents a holistic overview of the status quo in automated vehicle safety verification, validation and certification. In particular, the status and challenges of highly automated vehicle certification are pointed out, which should draw both academic and industrial attention as those challenges have a high reward as they are associated with significant pain points in automated vehicle deployment. We identified a critical missing link in quantifiable safety verification, a unified scenario coverage framework, and proposed such a definition to be expanded on by researchers of interest.

Formal safety verification is systematically over-viewed in this paper, including a state-of-the-art methodology review of its use in motion control safety verification and control synthesis with safety guarantees. As a form of formal methods, Reachability analysis has been used in multiple system control applications to guarantee robustness or constraint satisfaction. However, the lack of a unified and convenient engineering routine for safety specification penetration prevents the broader application of reachability analysis to more safety-critical functions in the industry. With emerging papers starting to bridge the formal temporal logic expressions with dynamical system control synthesis, we presented potential research opportunities and listed relevant tools.

ACKNOWLEDGMENT

The authors would like to thank Prof. Mo Chen at Simon Fraser University for providing high level feedback and suggestions.

⁷A more comprehensive description of this list is available at <https://github.com/nexthybrid/reachability-tools>

REFERENCES

- [1] J. E. Stellet, M. R. Zofka, J. Schumacher, T. Schamm, F. Niewels, and J. M. Zöllner, "Testing of advanced driver assistance towards automated driving: A survey and taxonomy on existing approaches and open questions," in *Proc. IEEE 18th Int. Conf. Intell. Transp. Syst.*, 2015, pp. 1455–1462.
- [2] D. Cao *et al.*, "Future directions of intelligent vehicles: Potentials, possibilities, and perspectives," *IEEE Trans. Intell. Veh.*, vol. 7, no. 1, pp. 7–10, Mar. 2022, doi: [10.1109/TIV.2022.3157049](https://doi.org/10.1109/TIV.2022.3157049).
- [3] S. Immen, "Kba erteilt erste genehmigung zum automatisierten fahren," Accessed: Jan. 10, 2022. [Online]. Available: https://www.kba.de/DE/Presse/Pressemitteilungen/Allgemein/2021/pm49_2021_erste_Genehmigung_automatisiertes_Fahren.html
- [4] Yahoo, "Honda wins world-first approval for level 3 autonomous car," Accessed: Jan. 10, 2022. [Online]. Available: <https://news.yahoo.com/honda-wins-world-first-approval-115115427.html>
- [5] T. Litman, *Autonomous Vehicle Implementation Predictions*. Canada: Victoria Victoria Transp. Policy Inst. Victoria, 2017.
- [6] S. S. Ho, "Complementary and competitive framing of driverless cars: Framing effects, attitude volatility, or attitude resistance," *Int. J. Public Opin. Res.*, vol. 33, no. 3, pp. 512–531, 2021.
- [7] J. Airbib and T. Seba, "Rethinking transportation 2020-2030: The disruption of transportation and the collapse of the internal-combustion vehicle and oil industries," *RethinkX Sector Disruption Rep.*, May 4, 2017.
- [8] E. Yurtsever, J. Lambert, A. Carballo, and K. Takeda, "A survey of autonomous driving: Common practices and emerging technologies," *IEEE Access*, vol. 8, pp. 58443–58469, 2020.
- [9] P. Koopman and M. Wagner, "Autonomous vehicle safety: An interdisciplinary challenge," *IEEE Intell. Transp. Syst. Mag.*, vol. 9, no. 1, pp. 90–96, Jan. 2017.
- [10] T. F. Kone, E. Bonjour, E. Levrat, F. Mayer, and S. Géronimi, "Safety demonstration of autonomous vehicles: A review and future research questions," in *Proc. Int. Conf. Complex Syst. Des. Manage.*, 2019, pp. 176–188.
- [11] J. Wang, L. Zhang, Y. Huang, and J. Zhao, "Safety of autonomous vehicles," *J. Adv. Transp.*, vol. 2020, Jun. 2020.
- [12] Z. Tahir and R. Alexander, "Coverage based testing for v&v and safety assurance of self-driving autonomous vehicles: A systematic literature review," in *Proc. IEEE Int. Conf. Artif. Intell. Testing*, 2020, pp. 23–30.
- [13] F. Batsch, S. Kanarachos, M. Cheah, R. Ponticelli, and M. Blundell, "A taxonomy of validation strategies to ensure the safe operation of highly automated vehicles," *J. Intell. Transp. Syst.*, vol. 26, no. 1, pp. 14–33, 2020.
- [14] S. Riedmaier, T. Ponn, D. Ludwig, B. Schick, and F. Diermeyer, "Survey on scenario-based safety assessment of automated vehicles," *IEEE Access*, vol. 8, pp. 87456–87477, 2020.
- [15] H. Kress-Gazit *et al.*, "Formalizing and guaranteeing human-robot interaction," *Commun. ACM*, vol. 64, no. 9, pp. 78–84, 2021.
- [16] J. Sun, H. Zhang, H. Zhou, R. Yu, and Y. Tian, "Scenario-based test automation for highly automated vehicles: A review and paving the way for systematic safety assurance," *IEEE Trans. Intell. Transp. Syst.*, to be published, doi: [10.1109/TITS.2021.3136353](https://doi.org/10.1109/TITS.2021.3136353).
- [17] W. Ding, C. Xu, M. Arief, H. Lin, B. Li, and D. Zhao, "A survey on safety-critical scenario generation for autonomous driving - A methodological perspective," 2022, *arXiv:2202.02215[cs.RO]*.
- [18] BMVI, "Germany will be the world leader in autonomous driving," Accessed: Oct. 01, 2022. [Online]. Available: <https://www.bmvi.de/SharedDocs/EN/Articles/DG/act-on-autonomous-driving.html>
- [19] U. N. E. C. for Europe, "Uniform provisions concerning the approval of vehicles with regard to automated lane keeping systems," Addendum 156 to the 1958 Agreement - UN Regulation No 157., pp. 34–36, 2021.
- [20] M. Chen, Q. Tam, S. C. Livingston, and M. Pavone, "Signal temporal logic meets Hamilton-Jacobi reachability: Connections and applications," in *Proc. Int. Workshop Algorithmic Found. Robot.*, 2018, pp. 581–601.
- [21] K. Leung *et al.*, "On infusing reachability-based safety assurance within planning frameworks for human-robot vehicle interactions," *Int. J. Robot. Res.*, vol. 39, no. 10–11, pp. 1326–1345, 2020.
- [22] B. Weng, L. J. Capito Ruiz, U. Ozguner, and K. Redmill, "Towards guaranteed safety assurance of automated driving systems with scenario sampling: An invariant set perspective," *IEEE Trans. Intell. Veh.*, to be published, doi: [10.1109/TIV.2021.3117049](https://doi.org/10.1109/TIV.2021.3117049).

- [23] *Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles*, SAE J3016_202104, SAE International, Warrendale, PA, USA, Apr. 30, 2021.
- [24] M. Wayland, "U.S. clears way for truly driverless vehicles without steering wheels," Accessed: Feb. 04, 2022. [Online]. Available: <https://www.cnn.com/2022/03/11/us-clears-way-for-driverless-vehicles-without-steering-wheels.html>
- [25] J. Capperella, "DMV approves nuro to use autonomous vehicles for commercial service," Accessed: Dec. 20, 2021. [Online]. Available: <https://www.dmv.ca.gov/portal/news-and-media/dmv-approves-nuro-to-use-autonomous-vehicles-for-commercial-service/>
- [26] DMV, "DMV approves cruise and waymo to use autonomous vehicles for commercial service in designated parts of bay area." Accessed: Jan. 01, 2022. [Online]. Available: <https://www.dmv.ca.gov/portal/news-and-media/117199-2/>
- [27] Auto2x, "Level 3-autonomous driving regulation finally coming after 3 years of delay," Accessed: Jan. 12, 2022. [Online]. Available: <https://auto2xtech.com/regulation-level-3-autonomous-driving-2021/>
- [28] S. Schmid, "Bundestag nimmt gesetz zum autonomen fahren an," Accessed: Jan. 10, 2022. [Online]. Available: <https://www.bundestag.de/dokumente/textarchiv/2021/kw20-de-autonomes-fahren-840196>
- [29] R. Bellan, "Germany gives greenlight to driverless vehicles on public roads," Accessed: Jan. 12, 2022. [Online]. Available: <https://techcrunch.com/2021/05/24/Germany-gives-greenlight-to-driverless-vehicles-on-public-roads/>
- [30] J. Capperella, "Mercedes drive pilot level 3 autonomous system to launch in Germany," Accessed: Jan. 12, 2022. [Online]. Available: <https://www.caranddriver.com/news/a38475565/mercedes-drive-pilot-autonomous-Germany/>
- [31] T. Stone, "Self-driving vehicles to become legal in UK in 2021," Accessed: Jan. 15, 2022. [Online]. Available: <https://www.traffictechnologytoday.com/news/autonomous-vehicles/self-driving-vehicles-to-become-legal-in-UK-in-2021.html>
- [32] D. Moss, "Self-driving cars could be legal by spring 2022," Accessed: Jan. 15, 2022. [Online]. Available: <https://www.whatcar.com/news/self-driving-cars-could-be-legal-by-spring-2022/n22965>
- [33] K. W. Mallesons, "China's legislation on autonomous cars rolls out," Accessed: Jan. 12, 2022. [Online]. Available: <https://www.chinalawinsight.com/2021/04/articles/corporate-ma/chinas-legislation-on-autonomous-cars-rolls-out/>
- [34] C. Feng, "China rolls out autonomous driving standards as carmakers work towards making self-driving a reality," Accessed: Jan. 12, 2022. [Online]. Available: <https://www.scmp.com/tech/tech-war/article/3149403/China-rolls-out-autonomous-driving-standards-carmakers-work-towards>
- [35] B. Goh, "Baidu, pony.ai approved for robotaxi services in beijing," Accessed: Jan. 12, 2022. [Online]. Available: <https://www.reuters.com/technology/baidu-ponyai-approved-robotaxi-services-beijing-2021-11-25/>
- [36] D. Roberts, J. Allsop, and P. Keeves, "Germany will be the world leader in autonomous driving," Accessed: Jan. 10, 2022. [Online]. Available: <https://hsfnotes.com/cav/2019/06/06/Japan-allows-level-3-automated-vehicles-on-public-roads/>
- [37] N. staff writers, "Japan to create legal framework for level 4 self-driving cars," Accessed: Jan. 10, 2022. [Online]. Available: <https://asia.nikkei.com/Business/Automobiles/Japan-to-create-legal-framework-for-level-4-self-driving-cars>
- [38] METI, "Demonstration test to be launched for evaluation of unmanned last-mile autonomous driving in wajima city to achieve mobility revolution," Accessed: Jan. 12, 2022. [Online]. Available: https://www.meti.go.jp/english/press/2017/1213_005.html
- [39] H.-y. Lee and Ji-yong; Lee, "Level 3 autonomous cars become legal in Korea from july," Accessed: Jan. 10, 2022. [Online]. Available: <https://pulsenews.co.kr/view.php?year=2020&no=15420>
- [40] J. Jung, "Korean government announces safety standards for level 3 automated vehicles," Accessed: Jan. 10, 2022. [Online]. Available: http://www.molit.go.kr/english/USR/BORD0201/m_28286/DTL.jsp?id=eng_mltm_new&mode=view&idx=2905
- [41] IIHS, "Advanced driver assistance: Autonomous vehicle laws," Accessed: Apr. 02, 2022. [Online]. Available: <https://www.iihs.org/topics/advanced-driver-assistance/autonomous-vehicle-laws#fn22>
- [42] P. Jin and Hyunjoo; Dave, "Waymo, cruise seek permits to charge for autonomous car rides in san francisco," Accessed: Jan. 12, 2022. [Online]. Available: <https://www.reuters.com/business/autos-transportation/exclusive-waymo-cruise-seek-permits-charge-self-driving-car-rides-san-francisco-2021-05-11/>
- [43] M. Subin and Samantha; Wayland, "Alphabet's waymo and gm's cruise get california DMV approval to run commercial autonomous car services," Accessed: Jan. 12, 2022. [Online]. Available: <https://www.cnn.com/2021/09/30/waymo-and-cruise-get-california-dmv-approval-to-run-driverless-cars.html>
- [44] D. Shephardson, "California issues permits to cruise, waymo for autonomous vehicle service," Accessed: Apr. 02, 2022. [Online]. Available: <https://www.reuters.com/technology/california-issues-permits-cruise-waymo-autonomous-vehicle-service-2022-02-28/>
- [45] M. Williams, "Prometheus-the european research programme for optimising the road transport system in Europe," in *Proc. IEE Colloq. Driver Inf.*, 1988, pp. 1–1.
- [46] U. Ozguner, C. Stiller, and K. Redmill, "Systems for safety and autonomous behavior in cars: The darpa grand challenge experience," *Proc. IEEE*, vol. 95, no. 2, pp. 397–412, 2007.
- [47] T. Cohen *et al.*, "A constructive role for social science in the development of automated vehicles," *Transp. Res. Interdiscipl. Perspectives*, vol. 6, 2020, Art. no. 100133.
- [48] N. H. T. S. Administration *et al.*, "Critical reasons for crashes investigated in the national motor vehicle crash causation survey," *NHTSA's Nat. Center Statist. Anal.*, Washington D.C., DOT-HS-812-115. Accessed: Jan. 12, 2022. [Online]. Available: <https://crashstats.nhtsa.dot.gov/Api/Public/Publication/812506>
- [49] N. Kalra and S. M. Paddock, "Driving to safety: How many miles of driving would it take to demonstrate autonomous vehicle reliability," *Transp. Res. Part A: Policy Pract.*, vol. 94, pp. 182–193, 2016.
- [50] F.-Y. Wang, "Metavehicles in the metaverse: Moving to a new phase for intelligent vehicles and smart mobility," *IEEE Trans. Intell. Veh.*, vol. 7, no. 1, pp. 1–5, Mar. 2022.
- [51] ISO, *Road Vehicles—Functional Safety—Part 1: Vocabulary*, ISO 26262-1:2018, International Organization for Standardization, Geneva, CH, Jan. 2018.
- [52] ISO, *Road vehicles—Safety of the Intended Functionality*, ISO/PAS 21448:2019, International Organization for Standardization, Geneva, CH, Jan. 2019.
- [53] C. DMV, "Disengagement reports," Accessed on : Dec. 06, 2021. [Online]. Available: <https://www.dmv.ca.gov/portal/vehicle-industry-services/autonomous-vehicles/disengagement-reports/>
- [54] Y. Kotoulas, "Mercedes to accept liability for accidents involving its new self driving tech," Accessed: Apr. 02, 2022. [Online]. Available: <https://www.insurancetimes.co.uk/news/mercedes-to-accept-liability-for-accidents-involving-its-new-self-driving-tech/1440701.article>
- [55] M. Grindal, J. Offutt, and S. F. Andler, "Combination testing strategies: A survey," *Softw. Testing, Verification Rel.*, vol. 15, no. 3, pp. 167–199, 2005.
- [56] C. Amersbach and H. Winner, "Defining required and feasible test coverage for scenario-based validation of highly automated vehicles," in *Proc. IEEE Intell. Transp. Syst. Conf.*, 2019, pp. 425–430.
- [57] D. Åsljung, J. Nilsson, and J. Fredriksson, "Comparing collision threat measures for verification of autonomous vehicles using extreme value theory," *IFAC-PapersOnLine*, vol. 49, no. 15, pp. 57–62, 2016.
- [58] P. Junietz, F. Bonakdar, B. Klamann, and H. Winner, "Criticality metric for the safety validation of automated driving using model predictive trajectory optimization," in *Proc. 21st Int. Conf. Intell. Transp. Syst.*, 2018, pp. 60–65.
- [59] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "On a formal model of safe and scalable self-driving cars," 2017, *arXiv:1708.06374[cs.Ro]*.
- [60] P. J. Antsaklis and A. N. Michel, *Linear Systems*, vol. 8. New York: Springer, 1997.
- [61] D. P. Bertsekas and I. B. Rhodes, "On the minimax reachability of target sets and target tubes," *Automatica*, vol. 7, no. 2, pp. 233–247, 1971.
- [62] M. S. Blumenthal, L. Fraade-Baran, R. Best, and J. L. Irwin, "Safe enough: Approaches to assessing acceptable safety for automated vehicles," *Tech. Rep. RR-A569-1*, RAND Corporation, 2020.
- [63] J. M. Wing, "A specifier's introduction to formal methods," *Computer*, vol. 23, no. 9, pp. 8–22, 1990.
- [64] A. Abate, M. Prandini, J. Lygeros, and S. Sastry, "Probabilistic reachability and safety for controlled discrete time stochastic hybrid systems," *Automatica*, vol. 44, no. 11, pp. 2724–2734, 2008.
- [65] C. Barrett and C. Tinelli, "Satisfiability modulo theories," in *Handbook of Model Checking*, pp. 305–343, New York: Springer, 2018.
- [66] F. Immler, "Verified reachability analysis of continuous systems," in *Proc. Int. Conf. Tools Algorithms Construction Anal. Syst.*, 2015, pp. 37–51.
- [67] E. M. Clarke, O. Grumberg, and D. E. Long, "Model checking and abstraction," *ACM Trans. Program. Lang. Syst.*, vol. 16, no. 5, pp. 1512–1542, 1994.

- [68] N. Kochdumper, P. Gassert, and M. Althoff, "Verification of collision avoidance for common road traffic scenarios," *EPiC Ser. Comput.*, vol. 80, pp. 184–194, 2021.
- [69] A. Corso, R. J. Moss, M. Koren, R. Lee, and M. J. Kochenderfer, "A survey of algorithms for black-box safety validation of cyber-physical systems," *J. Artif. Intell. Res.*, vol. 72, pp. 377–428, Oct. 2021.
- [70] P. Falcone, M. Ali, and J. Sjöberg, "Predictive threat assessment via reachability analysis and set invariance theory," *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 4, pp. 1352–1361, Dec. 2011.
- [71] J. Nilsson, J. Fredriksson, and A. C. Ödholm, "Verification of collision avoidance systems using reachability analysis," *IFAC Proc. Volumes*, vol. 47, no. 3, pp. 10676–10681, 2014.
- [72] R. Kianfar, P. Falcone, and J. Fredriksson, "Reachability analysis of cooperative adaptive cruise controller," in *Proc. 15th Int. IEEE Conf. Intell. Transp. Syst.*, 2012, pp. 1537–1542.
- [73] M. Althoff and J. M. Dolan, "Online verification of automated road vehicles using reachability analysis," *IEEE Trans. Robot.*, vol. 30, no. 4, pp. 903–918, Aug. 2014.
- [74] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "On a formal model of safe and scalable self-driving cars," 2018, *arXiv:1708.06374[cs.RO]*.
- [75] P. F. Orzechowski, K. Li, and M. Lauer, "Towards responsibility-sensitive safety of automated vehicles with reachable set analysis," in *Proc. IEEE Int. Conf. Connected Veh. Expo.*, 2019, pp. 1–6.
- [76] J. Yoo and R. Langari, "A predictive perception model and control strategy for collision-free autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 11, pp. 4078–4091, Nov. 2019.
- [77] A. Li, L. Sun, W. Zhan, M. Tomizuka, and M. Chen, "Prediction-based reachability for collision avoidance in autonomous driving," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 7908–7914.
- [78] S. Söntges and M. Althoff, "Determining the nonexistence of evasive trajectories for collision avoidance systems," in *Proc. IEEE 18th Int. Conf. Intell. Transp. Syst.*, 2015, pp. 956–961.
- [79] S. Söntges and M. Althoff, "Computing the drivable area of autonomous road vehicles in dynamic road scenes," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 6, pp. 1855–1866, Jun. 2018.
- [80] M. Althoff and J. M. Dolan, "Set-based computation of vehicle behaviors for the online verification of autonomous vehicles," in *Proc. 14th Int. IEEE Conf. Intell. Transp. Syst.*, 2011, pp. 1162–1167.
- [81] M. Althoff, O. Stursberg, and M. Buss, "Reachability analysis of linear systems with uncertain parameters and inputs," in *Proc. 46th IEEE Conf. Decis. Control*, 2007, pp. 726–732.
- [82] M. Althoff, O. Stursberg, and M. Buss, "Model-based probabilistic collision detection in autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 10, no. 2, pp. 299–310, Jun. 2009.
- [83] N. Isoda, K. Kogiso, and T. Asai, "Switching strategies of collision avoidance and tracking control for vehicles based on non-cooperative game and model predictive control," in *Proc. IEEE 22nd Int. Symp. Intell. Control*, 2007, pp. 178–183.
- [84] S. Vaskov, H. Larson, S. Kousik, M. Johnson-Roberson, and R. Vasudevan, "Not-at-fault driving in traffic: A reachability-based approach," in *Proc. IEEE Intell. Transp. Syst. Conf.*, 2019, pp. 2785–2790.
- [85] J. Nilsson, P. Falcone, and J. Vinter, "Safe transitions from automated to manual driving using driver controllability estimation," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 4, pp. 1806–1816, Aug. 2014.
- [86] S. Kousik, S. Vaskov, M. Johnson-Roberson, and R. Vasudevan, "Safe trajectory synthesis for autonomous driving in unforeseen environments," in *Proc. Dyn. Syst. Control Conf.*, vol. 58271, 2017, Art. no. V00 1T44A005.
- [87] A. Majumdar and R. Tedrake, "Funnel libraries for real-time robust feedback motion planning," *Int. J. Robot. Res.*, vol. 36, no. 8, pp. 947–982, 2017.
- [88] R. Ivanov, J. Weimer, R. Alur, G. J. Pappas, and I. Lee, "Verisig: Verifying safety properties of hybrid systems with neural network controllers," in *Proc. 22nd ACM Int. Conf. Hybrid Systems: Computation Control*, 2019, pp. 169–178.
- [89] R. Ivanov, T. J. Carpenter, J. Weimer, R. Alur, G. J. Pappas, and I. Lee, "Verifying the safety of autonomous systems with neural network controllers," *ACM Trans. Embedded Comput. Syst.*, vol. 20, no. 1, pp. 1–26, 2020.
- [90] X. Chen, E. Ábrahám, and S. Sankaranarayanan, "Flow*: An analyzer for non-linear hybrid systems," in *Proc. Int. Conf. Comput. Aided Verification*, 2013, pp. 258–263.
- [91] P. F. Orzechowski, A. Meyer, and M. Lauer, "Tackling occlusions and limited sensor range with set-based safety verification," in *Proc. 21st Int. Conf. Intell. Transp. Syst.*, 2018, pp. 1729–1736.
- [92] H.-D. Tran, L. V. Nguyen, P. Musau, W. Xiang, and T. T. Johnson, "Decentralized real-time safety verification for distributed cyber-physical systems," in *Proc. Int. Conf. Formal Techn. Distrib. Objects, Components, Syst.*, 2019, pp. 261–277.
- [93] L. Liebenwein *et al.*, "Compositional and contract-based verification for autonomous driving on road networks," in *Robotics Res.*, 2020, pp. 163–181.
- [94] P. Du *et al.*, "Online monitoring for safe pedestrian-vehicle interactions," in *Proc. IEEE 23rd Int. Conf. Intell. Transp. Syst.*, 2020, pp. 1–8.
- [95] T. Fraichard and H. Asama, "Inevitable collision states—a step towards safer robots," *Adv. Robot.*, vol. 18, no. 10, pp. 1001–1024, 2004.
- [96] S. Bouraine, T. Fraichard, and H. Salhi, "Relaxing the inevitable collision state concept to address provably safe mobile robot navigation with limited field-of-views in unknown dynamic environments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2011, pp. 2985–2991.
- [97] G. Savino, J. Brown, M. Rizzi, M. Pierini, and M. Fitzharris, "Triggering algorithm based on inevitable collision states for autonomous emergency braking (aeb) in motorcycle-to-car crashes," in *Proc. IEEE Intell. Veh. Symp. (IV)*, 2015, pp. 1195–1200.
- [98] L. Yu, R. Wang, and Z. Lu, "Autonomous emergency braking control based on inevitable collision state for multiple collision scenarios at intersection," in *Proc. IEEE Amer. Control Conf.*, 2021, pp. 148–153.
- [99] L. Martinez-Gomez and T. Fraichard, "Collision avoidance in dynamic environments: An ics-based solution and its comparative evaluation," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2009, pp. 100–105.
- [100] R. Parthasarathi and T. Fraichard, "An inevitable collision state-checker for a car-like vehicle," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2007, pp. 3068–3073.
- [101] L. Martinez-Gomez and T. Fraichard, "An efficient and generic 2d inevitable collision state-checker," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2008, pp. 234–241.
- [102] D. Althoff, M. Werling, N. Kaempchen, D. Wollherr, and M. Buss, "Lane-based safety assessment of road scenes using inevitable collision states," in *Proc. IEEE Intell. Veh. Symp.*, 2012, pp. 31–36.
- [103] A. Lawitzky, A. Nicklas, D. Wollherr, and M. Buss, "Determining states of inevitable collision using reachability analysis," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 4142–4147.
- [104] A. Bautin, L. Martinez-Gomez, and T. Fraichard, "Inevitable collision states: A probabilistic perspective," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2010, pp. 4022–4027.
- [105] D. Althoff, M. Althoff, D. Wollherr, and M. Buss, "Probabilistic collision state checker for crowded environments," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2010, pp. 1492–1498.
- [106] E. Frazzoli, M. A. Dahleh, and E. Feron, "Robust hybrid control for autonomous vehicle motion planning," in *Proc. 39th IEEE Conf. Decis. Control*, vol. 1, 2000, pp. 821–826.
- [107] E. Frazzoli, M. A. Dahleh, and E. Feron, "Real-time motion planning for agile autonomous vehicles," *J. Guid., Control, Dyn.*, vol. 25, no. 1, pp. 116–129, 2002.
- [108] A. Majumdar and R. Tedrake, "Robust online motion planning with regions of finite time invariance," in *Algorithmic Foundations Robotics X*, 2013, pp. 543–558.
- [109] A. J. Barry, P. R. Florence, and R. Tedrake, "High-speed autonomous obstacle avoidance with pushbroom stereo," *J. Field Robot.*, vol. 35, no. 1, pp. 52–68, 2018.
- [110] K. R. Williams *et al.*, "Trajectory planning with deep reinforcement learning in high-level action spaces," 2021, *arXiv:2110.00044[eeecs.SY]*.
- [111] Z. C. Goddard *et al.*, "Utilizing reinforcement learning to continuously improve a primitive-based motion planner," *J. Aerosp. Inf. Syst.*, pp. 1–12, 2021.
- [112] A. Bemporad and M. Morari, "Robust model predictive control: A survey," in *Robustness Identification Control*, 1999, pp. 207–226.
- [113] J. M. Bravo, T. Alamo, and E. F. Camacho, "Robust mpc of constrained discrete-time nonlinear systems based on approximated reachable sets," *Automatica*, vol. 42, no. 10, pp. 1745–1751, 2006.
- [114] S. Dixit *et al.*, "Trajectory planning for autonomous high-speed overtaking using mpc with terminal set constraints," in *Proc. IEEE 21st Int. Conf. Intell. Transp. Syst.*, 2018, pp. 1061–1068.
- [115] U. Rosolia and F. Borrelli, "Sample-based learning model predictive control for linear uncertain systems," in *Proc. IEEE 58th Conf. Decis. Control*, 2019, pp. 2702–2707.
- [116] M. Kvasnica, B. Takács, J. Holaza, and D. Ingole, "Reachability analysis and control synthesis for uncertain linear systems in mpt," *IFAC-PapersOnLine*, vol. 48, no. 14, pp. 302–307, 2015.

- [117] S. Kousik, S. Vaskov, F. Bu, M. Johnson-Roberson, and R. Vasudevan, "Bridging the gap between safety and real-time performance in receding-horizon trajectory design for mobile robots," *Int. J. Robot. Res.*, vol. 39, no. 12, pp. 1419–1469, 2020.
- [118] L. Hewing and M. N. Zeilinger, "Stochastic model predictive control for linear systems using probabilistic reachable sets," in *Proc. IEEE Conf. Decis. Control*, 2018, pp. 5182–5188.
- [119] L. Hewing and M. N. Zeilinger, "Scenario-based probabilistic reachable sets for recursively feasible stochastic model predictive control," *IEEE Contr. Syst. Lett.*, vol. 4, no. 2, pp. 450–455, Apr. 2020.
- [120] F. Gruber and M. Althoff, "Scalable robust model predictive control for linear sampled-data systems," in *Proc. IEEE 58th Conf. Decis. Control*, 2019, pp. 438–444.
- [121] D. Limon *et al.*, "Input-to-state stability: A unifying framework for robust model predictive control," in *Proc. Nonlinear Model Predictive Control*, 2009, pp. 1–26.
- [122] D. M. Raimondo, S. Rivero, C. N. Jones, and M. Morari, "A robust explicit nonlinear mpc controller with input-to-state stability guarantees," *IFAC Proc. Volumes*, vol. 44, no. 1, pp. 9284–9289, 2011.
- [123] A. Aswani, H. Gonzalez, S. S. Sastry, and C. Tomlin, "Provably safe and robust learning-based model predictive control," *Automatica*, vol. 49, no. 5, pp. 1216–1226, 2013.
- [124] M. Kvasnica, P. Bakaráč, and M. Klaučo, "Complexity reduction in explicit mpc: A reachability approach," *Syst. Control Lett.*, vol. 124, pp. 19–26, 2019.
- [125] S. Vaskov, U. Sharma, S. Kousik, M. Johnson-Roberson, and R. Vasudevan, "Guaranteed safe reachability-based trajectory design for a high-fidelity model of an autonomous passenger vehicle," in *Proc. IEEE Amer. Control Conf.*, 2019, pp. 705–710.
- [126] C. Hildebrandt, S. Elbaum, and N. Bezzo, "Blending kinematic and software models for tighter reachability analysis," in *Proc. IEEE/ACM 42nd Int. Conf. Softw. Engineering: New Ideas Emerg. Results*, 2020, pp. 33–36.
- [127] C. Huang, S. Xu, Z. Wang, S. Lan, W. Li, and Q. Zhu, "Opportunistic intermittent control with safety guarantees for autonomous systems," in *Proc. 57th ACM/IEEE Des. Automat. Conf.*, 2020, pp. 1–6.
- [128] A. Kozarev, J. Quindlen, J. How, and U. Topcu, "Case studies in data-driven verification of dynamical systems," in *Proc. 19th Int. Conf. Hybrid Systems: Computation Control*, 2016, pp. 81–86.
- [129] D. Fényes, B. Németh, and P. Gáspár, "Data-driven reachability analysis for the reconfiguration of vehicle control systems," *IFAC-PapersOnLine*, vol. 51, no. 24, pp. 831–836, 2018.
- [130] D. Fényes, B. Németh, and P. Gáspár, "Lpv-based autonomous vehicle control using the results of Big Data analysis on lateral dynamics," in *Proc. IEEE Amer. Control Conf.*, 2020, pp. 2250–2255.
- [131] K. Driggs-Campbell, R. Dong, and R. Bajcsy, "Robust, informative human-in-the-loop predictions via empirical reachable sets," *IEEE Trans. Intell. Veh.*, vol. 3, no. 3, pp. 300–309, Sep. 2018.
- [132] A. Alanwar, A. Koch, F. Allgöwer, and K. H. Johansson, "Data-driven reachability analysis using matrix zonotopes," in *Proc. Learn. Dyn. Control*, 2021, pp. 163–175.
- [133] M. J. Kochenderfer, J. E. Holland, and J. P. Chryssanthacopoulos, "Next-generation airborne collision avoidance system," *Lincoln Lab. J.*, vol. 19, no. 1, pp. 17–33, 2012.
- [134] K. D. Julian, M. J. Kochenderfer, and M. P. Owen, "Deep neural network compression for aircraft collision avoidance systems," *J. Guid., Control, Dyn.*, vol. 42, no. 3, pp. 598–608, 2019.
- [135] W. Xiang, H.-D. Tran, and T. T. Johnson, "Reachable set computation and safety verification for neural networks with relu activations," 2017, [arXiv:1712.08163](https://arxiv.org/abs/1712.08163).
- [136] H.-D. Tran *et al.*, "Parallelizable reachability analysis algorithms for feed-forward neural networks," in *Proc. IEEE/ACM 7th Int. Conf. Formal Methods Softw. Eng.*, 2019, pp. 51–60.
- [137] G. Yang, G. Qian, P. Lv, and H. Li, "Efficient verification of control systems with neural network controllers," in *Proc. 3rd Int. Conf. Vis., Image Signal Process.*, 2019, pp. 1–7.
- [138] L. K. Chung, A. Dai, D. Knowles, S. Kousik, and G. X. Gao, "Constrained feedforward neural network training via reachability analysis," 2021, [arXiv:2107.07696\[cs.LG\]](https://arxiv.org/abs/2107.07696).
- [139] H. Michael, F. Antonella, and W. Daniel, "Data-based reachability analysis for movement prediction of pedestrians and motion planning," in *Proc. IEEE Int. Conf. Veh. Electron. Saf.*, 2018, pp. 1–7.
- [140] S. Bansal, A. Bajcsy, E. Ratner, A. D. Dragan, and C. J. Tomlin, "A hamilton-jacobi reachability-based framework for predicting and analyzing human motion for safe planning," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2020, pp. 7149–7155.
- [141] J. F. Fisac *et al.*, "Probabilistically safe robot planning with confidence-based human predictions," 2018, [arXiv:1806.00109\[cs.RO\]](https://arxiv.org/abs/1806.00109).
- [142] M. Hartmann and D. Watznig, "Pedestrians walking on reachable sets and manifolds," in *Proc. IEEE Int. Conf. Mechatronics*, 2019, vol. 1, pp. 562–569.
- [143] F. Meng, J. Su, C. Liu, and W.-H. Chen, "Dynamic decision making in lane change: Game theory with receding horizon," in *Proc. IEEE UKACC 11th Int. Conf. Control*, 2016, pp. 1–6.
- [144] C. Huang, W. Li, and Q. Zhu, "Formal verification of weakly-hard systems," in *Proc. 22nd ACM Int. Conf. Hybrid Systems: Computation Control*, 2019, pp. 197–207.
- [145] M. Manassi, A. Liberman, A. Kosovicheva, K. Zhang, and D. Whitney, "Serial dependence in position occurs at the time of perception," *Psychon. Bull. Rev.*, vol. 25, no. 6, pp. 2245–2253, 2018.
- [146] J. Kolčák, J. Dubut, I. Hasuo, S.-y. Katsumata, D. Sprunger, and A. Yamada, "Relational differential dynamic logic," in *Proc. Int. Conf. Tools Algorithms Construction Anal. Syst.*, 2020, pp. 191–208.
- [147] A. Platzer, *Logical Foundations of Cyber-Physical Systems*, vol. 662. New York: Springer, 2018.
- [148] L. Dennis, M. Fisher, M. Slavkovik, and M. Webster, "Formal verification of ethical choices in autonomous systems," *Robot. Auton. Syst.*, vol. 77, pp. 1–14, 2016.
- [149] L. Dennis, M. Fisher, M. Slavkovik, and M. Webster, "Ethical choice in unforeseen circumstances," in *Proc. Conf. towards Auton. robotic Syst.*, 2013, pp. 433–445.
- [150] R. Hilpinen, *Deontic Logic: Introductory and Systematic Readings*, vol. 33. New York: Springer Science & Business Media, 2012.
- [151] A. Rizaldi and M. Althoff, "Formalising traffic rules for accountability of autonomous vehicles," in *Proc. IEEE 18th Int. Conf. Intell. Transp. Syst.*, 2015, pp. 1658–1665.
- [152] A. Bajcsy, S. Bansal, E. Bronstein, V. Tolani, and C. J. Tomlin, "An efficient reachability-based framework for provably safe autonomous navigation in unknown environments," in *Proc. IEEE 58th Conf. Decis. Control*, 2019, pp. 1758–1765.
- [153] S. L. Herbert, S. Bansal, S. Ghosh, and C. J. Tomlin, "Reachability-based safety guarantees using efficient initializations," in *Proc. IEEE 58th Conf. Decis. Control*, 2019, pp. 4810–4816.
- [154] E. Yel and N. Bezzo, "Fast run-time monitoring, replanning, and recovery for safe autonomous system operations," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2019, pp. 1661–1667.
- [155] M. Althoff and S. Magdici, "Set-based prediction of traffic participants on arbitrary road networks," *IEEE Trans. Intell. Veh.*, vol. 1, no. 2, pp. 187–202, Jun. 2016.
- [156] W. Xiang, H.-D. Tran, J. A. Rosenfeld, and T. T. Johnson, "Reachable set estimation and safety verification for piecewise linear systems with neural network controllers," in *Proc. IEEE Annu. Amer. Control Conf.*, 2018, pp. 1574–1579.
- [157] A. Girard, "Reachability of uncertain linear systems using zonotopes," in *Hybrid Systems: Computation Control* (M. Morari and L. Thiele, eds.), 2005, pp. 291–305.
- [158] H.-D. Tran, F. Cai, M. L. Diego, P. Musau, T. T. Johnson, and X. Koutsoukos, "Safety verification of cyber-physical systems with reinforcement learning control," *ACM Trans. Embedded Comput. Syst.*, vol. 18, no. 5s, pp. 1–22, 2019.
- [159] E. Asarin, T. Dang, and A. Girard, "Reachability analysis of nonlinear systems using conservative approximation," in *Hybrid Systems: Computation Control* (O. Maler and A. Pnueli, eds.), (Berlin, Heidelberg), 2003, pp. 20–35.
- [160] M. Althoff, D. Althoff, D. Wollherr, and M. Buss, "Safety verification of autonomous vehicles for coordinated evasive maneuvers," in *Proc. IEEE Intell. Veh. Symp.*, 2010, pp. 1078–1083.
- [161] H. A. Hansen, G. Schneider, and M. Steffen, "Reachability analysis of non-linear planar autonomous systems," in *Proc. Int. Conf. Fundamentals Softw. Eng.*, 2011, pp. 206–220.
- [162] S. Bak, S. Bogomolov, P. S. Duggirala, A. R. Gerlach, and K. Potomkin, "Reachability of black-box nonlinear systems after koopman operator linearization," 2021, [arXiv:2105.00886](https://arxiv.org/abs/2105.00886).
- [163] L. Liebenwein, C. Baykal, I. Gilitichenski, S. Karaman, and D. Rus, "Sampling-based approximation algorithms for reachability analysis with provable guarantees," in *Proc. Robotics: Sci. Syst.*, Pittsburgh, Pennsylvania, Jun. 26–28, 2018.
- [164] T. T. Johnson, "Arch-comp20 repeatability evaluation report," in *ARCH20. 7th Int. Workshop Appl. Verification Continuous Hybrid Syst.* (G. Frehse and M. Althoff, eds.), 2020, vol. 74, pp. 175–183.
- [165] M. Althoff, O. Stursberg, and M. Buss, "Reachability analysis of nonlinear systems with uncertain parameters using conservative linearization," in *Proc. 47th IEEE Conf. Decis. Control*, 2008, pp. 4042–4048.

- [166] P. Flüs and O. Stursberg, “Online control of affine systems in stochastically modeled contexts,” *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 3354–3360, 2020.
- [167] C. Fan, B. Qi, S. Mitra, and M. Viswanathan, “D ry vr: Data-driven verification and compositional reasoning for automotive systems,” in *Proc. Int. Conf. Comput. Aided Verification*, 2017, pp. 441–461.
- [168] X. Chen, S. Schupp, I. B. Makhlof, E. Ábrahám, G. Frehse, and S. Kowalewski, “A benchmark suite for hybrid systems reachability analysis,” in *Proc. NASA Formal Methods Symp.*, 2015, pp. 408–414.



Tong Zhao (Member, IEEE) received the B.S. degrees in physics from Nanjing University, Nanjing, China, and the University of Waterloo, Waterloo, ON, Canada, in 2012, and the M.S. degree in electrical engineering from Wuhan University, Wuhan, China, in 2016. He is currently working toward the Ph.D. degree in mechanical engineering with Ohio State University, Columbus, OH, USA.

In 2019, he was a Control Algorithm Intern with tuSimple, Tucson, AZ. In 2020 and 2021, he was a Software Engineer Intern with Mathworks. His research interests include limit-handling vehicle control algorithm and AV control safety guarantee through formal methods.



Ekim Yurtsever (Member, IEEE) received the B.S. and M.S. degrees from Istanbul Technical University, Istanbul, Turkey, in 2012 and 2014 respectively, and the Ph.D. degree in information science from Nagoya University, Nagoya, Japan, in 2019. Since 2019, he has been with the Department of Electrical and Computer Engineering, The Ohio State University, Columbus, OH, USA, as a Research Associate.

His research interests include artificial intelligence, machine learning, computer vision, reinforcement learning, intelligent transportation systems, and au-

tomated driving systems.



Joel A. Paulson received the B.S. degree from the University of Texas Austin, Austin, TX, USA, in 2011, the M.S.CEP and the Ph.D. degree in chemical engineering from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 2013 and 2016, respectively.

From 2016 to 2019, he was a Postdoc Scholar with the Department of Chemical and Biomolecular Engineering, University of California, Berkeley, Berkeley, CA, USA. Since 2019, he has been an Assistant Professor with the Department of Chemical and Biomolecular Engineering, Ohio State University, Columbus, OH, USA. His research interests include development of optimization, machine learning, and multi-scale simulation methods to improve the quality, efficiency, and sustainability of engineered products and processes. Two particular areas of focus in his group are: (i) construction of predictive models using a combination of physics-based equations and data and (ii) the formulation of real-time decision-support tasks under uncertainty as structured optimization problems that are solved efficiently using state-of-the-art algorithms. The developed strategies have been successfully applied to a broad range of systems including continuous pharmaceutical manufacturing, colloidal self-assembly, and non-equilibrium plasma jets.



Giorgio Rizzoni (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in electronics & communication engineering from the University of Michigan, Ann Arbor, MI, USA, in 1980, 1982, and 1986, respectively.

Since 1999, he has been the Director of the Ohio State University, Columbus, OH, USA, Center for Automotive Research (CAR), an interdisciplinary university research center in the OSU College of Engineering. His research interests include modeling, control and diagnosis of advanced propulsion systems, vehicle fault diagnosis and prognosis, electrified powertrains and energy storage systems, vehicle safety and intelligence, and sustainable mobility. He has contributed to the development of graduate curricula in these areas, and was the Director of three U.S. Department of Energy Graduate Automotive Technology Education Centers of Excellence: Hybrid Drivetrains and Control Systems (1998–2004), Advanced Propulsion Systems (2005–2011), and Energy Efficient Vehicles for Sustainable Mobility (2011–2016). Between 2011 and 2016 he was the OSU Site Director for the U.S. Department of Energy China-USA Clean Energy Research Center - Clean Vehicles. He is currently leading an ARPA-E project in the NEXTCAR program. During his career with Ohio State, Prof. Rizzoni has directed externally sponsored research projects funded by major government agencies and by the automotive industry in approximately equal proportion. Prof. Rizzoni is a Fellow of SAE (2005), a Fellow of IEEE (2004). He was the recipient of the 1991 National Science Foundation Presidential Young Investigator Award, and of many other technical and teaching awards.